How Much Does Early Warning Improve Famine Response?

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Abstract

The U.S.-funded Famine Early Warning Systems Network (FEWS NET) forecasts food crises in about 30 countries, at a cost of about $45 million a year. After the 2025 aid freeze it moved to the State Department, which must now decide whether to keep paying for it. We score its 2011–2024 forecasts against independent assessments by the Integrated Food Security Phase Classification and the Cadre Harmonisé, and link them to humanitarian funding. Four to eight months ahead, about 54 percent of forecast emergencies did not materialize (Type I error), and the system attributes 56 percent of those to humanitarian aid; over the same horizon it missed 53 percent of new emergencies (Type II error), though it had forecast at least a crisis for 89 percent of them. Added to public data on rainfall, conflict and prices, the forecasts improve predictions of which areas will tip into crisis: given one area that does and one that does not, they rank the first higher 85 percent of the time, against 79 percent with public data alone. Donors have historically increased food aid to these countries when the system's forecasts said more than public data did, and cut it when they said less; on net, this accounts for 7–9 percent of food aid, or $274–370 million a year, which we treat as an upper bound on the system's influence on the size of the budget. Holding the food-aid budget fixed, placing aid by the forecasts rather than by public information would make it 4–8 percent more effective at reaching people in crisis; extrapolated to all covered countries, that is roughly 3.9–4.2 million person-months a year kept out of crisis, worth 2.0–2.1 times the system's cost in food aid.

JEL codes: O19, Q18, F35, I15. **Keywords:** famine, early warning, humanitarian aid, forecasting, food security.

*This paper is fully AI generated. Claude Opus 5.5 assembled the data, wrote the code, ran the analysis and drafted the text. Human direction consisted of 88 prompts totaling about 5,100 words between 30 September and 14 October 2026. Replication code is available at `https://github.com/justinsandefur/fewsnet-evaluation`.

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1 Introduction

In August 2010, the Famine Early Warning Systems Network warned that failing rains would push southern Somalia into crisis. By March 2011 it was warning of possible famine. Famine was declared in July 2011, and an estimated 258,000 people died between 2010 and 2012, about half of them young children (Salama et al., 2012; Checchi and Robinson, 2013). The post-mortems agreed that the warnings had been timely and right; the response was not (Hillbruner and Moloney, 2012). Fourteen years later the question has flipped. In late January 2025, during the freeze on U.S. foreign assistance, FEWS NET went offline. It had been publishing maps and forecasts for 32 countries; it came back in stages from August 2025, now under the State Department, which must decide whether to keep funding it (Devex, 2025; NPR, 2026). Somalia 2011 shows that a warning system can be right and still not save lives. The question now is whether it is worth paying for, and that is the question this paper is written to inform.

This paper asks what the system's warnings are worth, using the data it publishes itself and the independent assessments that run alongside it. We organize the evaluation around the two ways a warning system can fail. A *Type II error* is a missed emergency: a crisis that arrives without warning. A *Type I error* is a false alarm: a warned crisis that never comes. The two errors are not symmetric in what they reveal. If warnings work — if they bring aid that prevents the crisis — a correct warning looks like a false alarm. Aid can turn a true positive into an apparent Type I error, but it cannot create a Type II error. So observed false alarms overstate how often the forecasts were wrong, while observed misses are a lower bound on how often they failed to see a crisis coming. This asymmetry, and the system's own judgments about where aid is holding back a crisis, shape the whole analysis.

We combine four data sources: (i) every FEWS NET current-situation map and medium-term forecast for 16,684 areas in 37 countries, 2011–2024; (ii) the Cadre Harmonisé in West Africa and the multi-agency Integrated Food Security Phase Classification (IPC) elsewhere, which publish their own area classifications and projections; (iii) humanitarian funding commitments from OCHA's Financial Tracking Service; and (iv) local rainfall, political violence and food prices — the raw signals a donor could watch without it.

Our first findings concern the two errors, scored on forecasts made four to eight months ahead, the window that matters for mobilizing aid. About 54 percent of forecast emergencies (Phase 4 on the five-phase IPC scale) did not materialize (Type I error). By FEWS NET's own estimate, 56 percent of these were held back by humanitarian aid, which would be a sign of the system working, though nearly half of the areas it credits to aid were already flagged before the warning, and we find no sign that warnings come true less often when funding then rises unusually fast. Over the same horizon, the forecasts put only 47 percent of areas that newly entered Emergency in Emergency, missing 53 percent (Type II error). Most misses were errors of degree: 89 percent of new emergencies had been forecast to be at least in Crisis, and only 4 percent arrived with no warning of Crisis. The warnings carry real signal, but they mostly arrive late and one phase too low. Famine is too rare for error rates: there were four episodes, and in none did the most likely

forecast four to eight months ahead show famine, though the written reports raised the risk of famine six to twelve months before it was declared (Section 5.1).

Second, the system knows things that other sources do not. We build two benchmarks for what a donor could know without it: one from rainfall, conflict and food prices, and one that adds the latest official IPC or Cadre Harmonisé analysis. Without it, donors would usually have had only the first: just 25 percent of the areas and rounds it maps had an official analysis from the previous eight months. Scored against the next independent assessment, adding its maps and forecasts improves the ranking of which areas will tip into Crisis: given one area that does and one that does not, the first is ranked higher 85 percent of the time, against 79 percent with public data alone. Over the benchmark with official analyses the gain is smaller (0.84 to 0.86 on the same scale). In a head-to-head with the official projections for the same areas and seasons, its forecasts catch somewhat fewer new crises (37 versus 42 percent) but raises fewer false alarms (7 versus 12 percent), and its forecast predicts the next official analysis even after controlling for the official projection.

Third, money follows the system's distinctive information. Across countries, humanitarian funding tracks its maps closely. Within a country over time, funding follows its forecasts rather than its current maps. Splitting the forecast into the part predictable from the country, the season, rainfall, conflict and prices, and FEWS NET's own contribution, only the latter predicts funding: a 10 percentage point rise in the share of a country's areas that FEWS NET forecasts to be in Crisis, beyond what that information predicts, comes with 12 percent more humanitarian funding and 33 percent more food and nutrition funding over the next six months. A partial check within Somalia and Sudan, using the small share of aid that can be located by district, finds no sign that located aid follows FEWS NET's own information, though that slice of aid is too small to settle the question.

Fourth, we value the forecasts on two margins, in person-months of food insecurity, the outcome the system is designed to forecast; its phases are a much weaker guide to malnutrition and mortality. The first is potential: holding the food-aid budget fixed, how much better could a planner place it with the forecasts than with public information alone? With the budget set to what actual food funding covers and realistic targeting within areas, the forecasts raise the share of rations reaching people who would otherwise be in Crisis by 4–8 percent, keeping 2.4–2.6 million more person-months out of Crisis a year in our sample of 24 countries, or 3.9–4.2 million extrapolated to all FEWS NET countries. At the cost of a ration, that is worth $88–95 million of food aid a year, 2.0–2.1 times the system's budget (1.2–1.3 times in the sample alone). The second margin is historical influence on the size of the budget: a net 7–9 percent of food security and nutrition funding, $274–370 million a year, moved with its distinctive information. That is an upper bound, since some of the money would have gone to other crises anyway and the funding response is an association.

Fifth, the value lies in the analysts' judgment, not in data a funder could buy separately. A pilot that turns 502 of the system's written reports into data, using a large language model, finds

that the text adds almost nothing to its maps, and market prices add little to forecasts, from WFP or from the system's own monitoring.

These results do not show what FEWS NET has caused. The allocation value is what FEWS NET's information could achieve if used well; the funding response is an association. The obvious natural experiments — the 2025 shutdown and earlier changes in FEWS NET coverage — turn out to be too weak to identify effects on welfare (Section 9). What the data do show is that FEWS NET produces information that other sources lack, that this information is borne out by later independent assessments, and that humanitarian money moves with it.

Related literature. Several studies score FEWS NET's forecasts against its own later maps. They find headline accuracy of 78–92 percent, falling in pastoral and conflict-affected areas and in strong El Niño years (Choularton and Krishnamurthy, 2019; Krishnamurthy et al., 2020; Backer and Billing, 2021; Bertetti et al., 2024). Because food security changes slowly, a forecast that nothing will change does nearly as well: Bertetti et al. (2024) find 78 percent accuracy for FEWS NET's near-term forecasts against 76 percent for persistence. Andrée et al. (2020) show that FEWS NET's outlooks miss most transitions into Crisis while rarely raising false alarms, and that machine-learning models trained on public data can trade some of that conservatism for more detection. Outside FEWS NET, Lentz et al. (2019) find that IPC zone classifications explain almost none of the variation in household food security measured by surveys in Malawi, and Lentz et al. (2025) argue that official IPC figures undercount acute hunger by about a fifth. Seyffer et al. (2026) compare FEWS NET and IPC projections in Somalia and find similar agreement with later assessments. We add three things. First, we score FEWS NET against independent assessments, not only its own later maps, and compare it directly with the official projections. Second, we use the Type I/Type II framework and FEWS NET's own aid flag to separate false alarms from crises held back by aid. Third, we measure FEWS NET's value relative to what donors would otherwise know, and link that marginal information to humanitarian funding. The last step connects to work on what drives disaster and humanitarian aid, which finds that need explains only part of the allocation, alongside news coverage and donor politics (Eisensee and Strömberg, 2007; Drury et al., 2005; Fink and Redaelli, 2011).

The rest of the paper runs as follows. Section 2 describes FEWS NET and sets out why aid clouds the measurement of both errors. Section 3 describes the data. Sections 4 and 5 measure missed emergencies and false alarms. Section 6 asks what FEWS NET adds to what donors could otherwise know, and Section 7 whether money follows it. Section 8 values FEWS NET on two margins. Section 9 explains what the data cannot tell us, and Section 10 concludes.

2 FEWS NET and the measurement problem

2.1 What FEWS NET produces

USAID created FEWS NET in 1985, after the Ethiopian and Sudanese famines of the mid-1980s. It monitors rainfall, harvests, markets, livestock, conflict and humanitarian assistance, and turns them into maps that classify each area of a country on the five-phase scale of the Integrated Food Security Phase Classification (IPC): Minimal (1), Stressed (2), Crisis (3), Emergency (4) and Famine (5) (IPC Global Partners, 2021). An area is placed in the highest phase that describes at least 20 percent of its population.

Three features of FEWS NET's products matter for this paper.

- **Current maps and forecasts.** Each round, FEWS NET publishes a map of the current situation and two forecasts: a near-term one for roughly the next four months and a medium-term one for the four months after that. Since 2016 these rounds have come three times a year, in February, June and October; before 2016, quarterly (FEWS NET, 2018).
- **Forecasts assume aid.** The forecasts build in humanitarian assistance that is planned and likely to be funded. A forecast of Emergency therefore means "Emergency even with the aid we expect," not "Emergency if nobody acts."
- **The aid flag.** FEWS NET marks an area with "!" when it judges that humanitarian assistance is keeping the area at least one phase better than it would otherwise be. The flag is FEWS NET's own estimate of what aid is preventing.

FEWS NET is not the only source of classifications. The IPC, a partnership of UN agencies, NGOs and governments, publishes area classifications and projections in many countries, and the Cadre Harmonisé does the same in West Africa. FEWS NET staff take part in many of these analyses. In much of FEWS NET's coverage, though, no official analysis exists for months at a time.

2.2 Two errors, and why aid hides them

Consider an area with underlying need N — the phase it would reach without aid — and observed phase P after any aid arrives. A forecaster issues warning W about N; donors respond with aid A, which lowers P relative to N. A Type II error is a crisis ($P \geq 4$, say) without a prior warning; a Type I error is a warning followed by no crisis.

Aid treats the two errors differently. Where a warning brings aid that works, $P < N$: a correct warning is recorded as a false alarm. Where no warning was issued, aid is less likely, and P is closer to N. So:

- observed false alarms overstate the true Type I error rate, by the number of crises prevented; and

- observed misses understate the true Type II error rate only if unwarned crises were nonetheless prevented by aid, which is less likely. Observed misses are close to a lower bound.

The same logic means that a system whose warnings work will look less accurate than one whose warnings are ignored. Any evaluation that scores forecasts only against outcomes penalizes successful early warning. This is the familiar problem of judging any response that goes where the problems are — hospitals have sicker patients, and countries under IMF programs have worse economies (Barro and Lee, 2005).

FEWS NET's aid flag offers a partial way out. If we take it at face value, an area mapped as Crisis with the flag had an underlying need of Emergency, and we can score warnings against need (P plus one if flagged) rather than outcome. We use this throughout, with two caveats: the same analysts produce the warning and the flag, and the flag records ongoing aid, not only aid that responded to the warning.

2.3 Two examples

Figures 1 and 2 show what FEWS NET's products look like in two crises. In each, the left panels show signals any donor could watch, and the right panels show FEWS NET's forecast and what it later mapped.

Somalia, 2016–17. The October–December 2016 rains failed: across Somalia's 199 FEWS NET areas, rainfall over the six months to January 2017 averaged −0.8 standard deviations below normal, with the deepest deficits in the north (panel A). Political violence was concentrated in the south (panel B). In February 2017 FEWS NET forecast Emergency for 26 areas by June, up from 16 at the time, including much of the north (panel C). By June, 29 areas were in Emergency — mostly in the south — and the north was mapped one phase lower, in Crisis, but flagged as held back by aid (panel D). In all, FEWS NET flagged 95 of the 199 areas (48 percent). This is the Type I ambiguity in miniature. The northern warnings did not come true as forecast, and FEWS NET's own map says that is because aid arrived.

Ethiopia, 2015–16. The El Niño drought hit the northern and central highlands: rainfall over the six months to September 2015 averaged −1.1 standard deviations below normal across Ethiopia's 678 FEWS NET areas, and much more in the worst-hit zones (panel A). Conflict was low and plays no role here. In October 2015, FEWS NET forecast Emergency for 26 areas by February 2016, up from 9 (panels B and C). By February, 24 areas were in Emergency, almost exactly where forecast, and FEWS NET flagged 103 areas (15 percent) as held back by aid. Many flags fell in areas mapped as Crisis in the north and as Stressed or Minimal in the south (panel D). Here the forecast came true, and FEWS NET credits aid with keeping the surrounding areas from following.

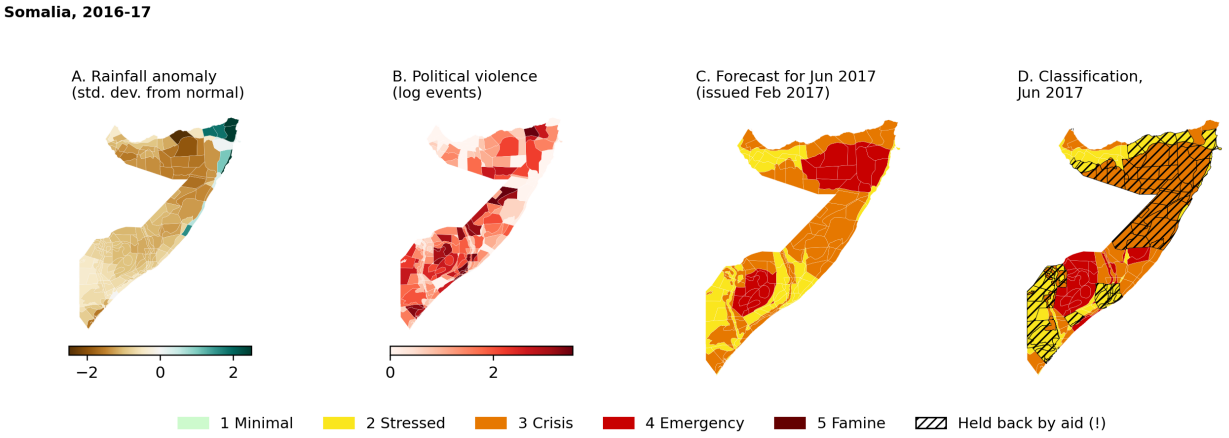


Figure 1: Somalia, 2016–17. FEWS NET areas. A: rainfall over the six months to January 2017, in standard deviations from the same months in the previous 20 years (CHIRPS). B: political violence events in the six months to January 2017 (ACLED, log of one plus events). C: FEWS NET's forecast, issued February 2017, for June 2017. D: FEWS NET's classification for June 2017; hatching marks areas FEWS NET flagged as held at least one phase better by humanitarian assistance.

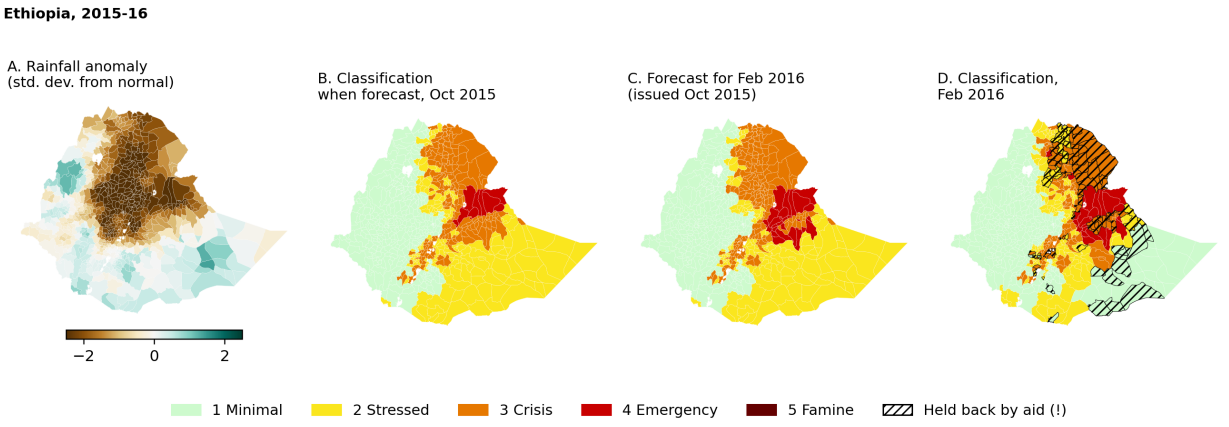


Figure 2: Ethiopia, 2015–16. FEWS NET areas. A: rainfall over the six months to September 2015, in standard deviations from the same months in the previous 20 years (CHIRPS). B: FEWS NET's classification in October 2015, when the forecast was issued. C: FEWS NET's forecast, issued October 2015, for February 2016. D: FEWS NET's classification for February 2016; hatching marks areas FEWS NET flagged as held at least one phase better by humanitarian assistance.

3 Data

Table 1 summarizes the data. All sources are public; Appendix A gives details and match rates.

FEWS NET. We use every current-situation map and forecast published through FEWS NET's data warehouse from January 2011 to December 2024 — 209,404 area-months in 16,684 areas and 37 countries. We stop at 2024 so that the 2025 shutdown does not distort the sample.

Independent classifications. The Cadre Harmonisé covers 19 West and Central African countries from 2014 and the IPC some 30 countries from 2017. For each area and analysis, we record whether the area is classified in Crisis or worse and the share of its population in Crisis or worse, for both the current situation and the projection. We overlay these areas on FEWS NET's to compare like with like. These classifications are independent of FEWS NET's own maps, but not of its staff, who join many IPC and Cadre Harmonisé analyses. They are the best independent benchmark available, not a perfect one.

Funding. The Financial Tracking Service records humanitarian commitments by recipient country and date. Over 2011–2024, FEWS NET countries received about 56 percent of all humanitarian funding; the United States provided about 52 percent of food and nutrition funding to these countries. We sum commitments to each country over the six months after each FEWS NET map round.

Raw signals. To represent what a donor could know without FEWS NET, we measure, for each FEWS NET area and round: rainfall over the previous 6 and 12 months relative to the same months in the preceding 20 years (CHIRPS); political violence events and deaths in the previous 6 months and their change on a year earlier (ACLED), apportioned from districts by area; and 3- and 12-month changes in retail cereal prices at the nearest WFP market. Rainfall covers 99 percent of area-rounds, conflict 48 percent and prices 28 percent; models treat missing values as missing.

Coverage. Figure 3 shows why FEWS NET matters. Over 2016–2024, the countries FEWS NET monitored accounted for 69 percent of the world's person-years in Crisis or worse, and 82 percent of those in Emergency or worse, across the 70 countries in the Global Report on Food Crises. The largest crises outside its coverage were Syria, Pakistan, Bangladesh and Myanmar.

Table 1: Data sources

Source	Content	Coverage used
FEWS NET Data Warehouse (FEWS NET, 2026)	Current-situation phases, near- and medium-term forecasts and aid flags for each FEWS NET area	209,404 area-months; 16,684 areas in 37 countries; 2011–2024
Cadre Harmonisé (Cadre Harmonisé, 2026)	Area classifications and projections, West and Central Africa	2014–2026; district or region level
IPC (IPC Global Partners, 2026)	Area classifications and projections outside West Africa	2017–2026
Global Report on Food Crises (FSIN and Global Network Against Food Crises, 2025)	People in Crisis or worse, by country and year	2016–2024; 70 countries
OCHA Financial Tracking Service (UN OCHA, 2026)	Humanitarian funding commitments by recipient, sector and date	2011–2024
CHIRPS (Funk et al., 2015)	Monthly rainfall, 0.05° grid (aggregated to 0.2°)	1981–2026
ACLED (Raleigh et al., 2010; ACLED, 2026)	Political violence events and deaths by district and month	19 countries with sub-national data
WFP market prices (World Food Programme, 2026)	Retail prices of cereals and tubers by market	Nearest market within 150 km

Notes: FEWS NET areas are livelihood zones or administrative units, depending on the country. IPC and Cadre Harmonisé areas are matched to FEWS NET areas by spatial overlay.

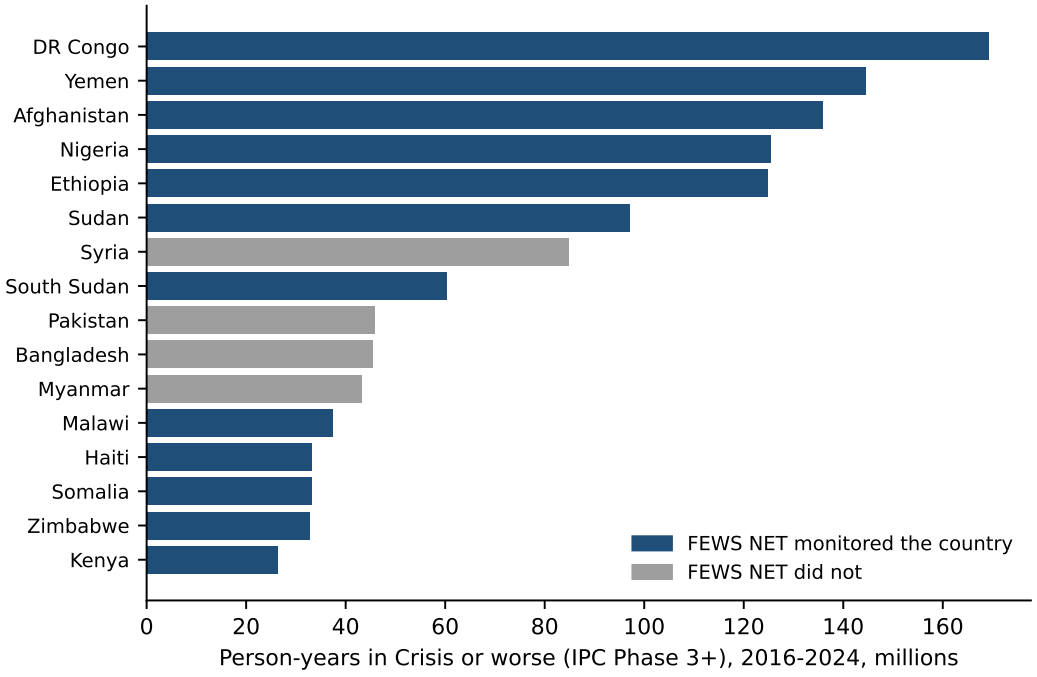


Figure 3: Where the world's food crises are. Person-years in Crisis or worse (IPC Phase 3 and above), 2016–2024, for the 16 countries with the most, by whether FEWS NET monitored the country. Source: Global Report on Food Crises country-year database.

4 Type II errors: missed emergencies

We define a new emergency as an area mapped in Emergency or worse whose previous map, no more than six months earlier, placed it below Emergency. Between 2011 and 2024 there were 1,562 such cases. For each, we look back at every forecast FEWS NET issued for that area and month in the preceding one to eight months.

Figure 4 shows the result. The thick line plots, by how far ahead the forecast was made, the share of new emergencies that FEWS NET forecast to be in Emergency; the dashed line, the share forecast to be in Crisis or worse. Behind the line, each faint dot is all the areas in one country whose Emergency began in the same month (93 such country-months; dot size is the number of areas). Almost all new emergencies were foreseen as crises: at four months, 95 percent had been forecast to be in at least Crisis. Few were foreseen as emergencies: 22 percent at eight months, 42 percent at four months and 63 percent one month ahead. Of the dots that cover more than one area, 54 percent sit at exactly 0 or 100 percent: FEWS NET tends to call the areas in a country-month together, so there are far fewer independent warnings than the 1,562 areas suggest. Taking the most severe forecast made four to eight months ahead — the window that matters for mobilizing aid — 47 percent were forecast as Emergency, 42 percent as Crisis, and 11 percent as Stressed or better.

These numbers say that FEWS NET's Type II errors are mostly errors of degree, not of kind. Only 4 percent of new emergencies came with no warning of at least Crisis. The warnings also discriminate: four months ahead, FEWS NET forecast Emergency for 42 percent of areas that went on to tip into Emergency, but for only 1.3 percent (on average across horizons) of comparable areas that did not. What FEWS NET does badly is forecast escalation from Crisis to Emergency — the step at which, in the IPC's definitions, very high acute malnutrition and excess mortality are expected.

A wider lens tells the same story. Food security changes slowly: in 79 percent of area-rounds, an area's next map shows the same phase as its current one. FEWS NET's medium-term forecast matches the next map 84 percent of the time, against 79 percent for a forecast that nothing changes. Among the 21 percent of area-rounds in which the phase did change, the forecast was right 55 percent of the time. These numbers score FEWS NET against itself and so flatter it, but the conclusion is consistent with earlier evaluations: most of FEWS NET's headline accuracy comes from stable areas, and its skill on change is real but partial.

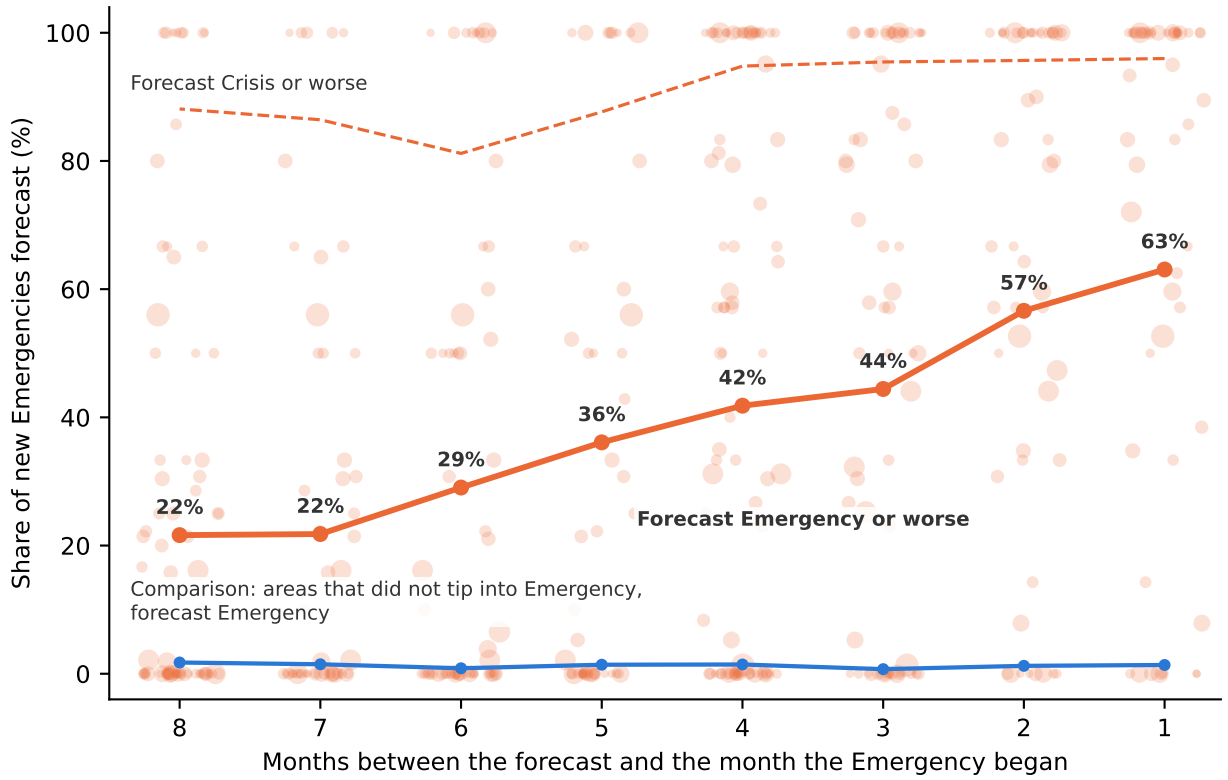


Figure 4: Were new emergencies foreseen? 1,562 FEWS NET areas that entered Emergency or worse (IPC Phase 4+) in 2011–2024, after a previous map below Emergency. Thick orange line: share of these areas that FEWS NET had forecast, k months earlier, to be in Emergency or worse for the outcome month or the three months before it; dashed line: forecast in Crisis or worse. Faint dots: all the areas in one country whose Emergency began in the same month (93 country-months; size = number of areas; spread sideways so they do not overlap). Blue line: share of areas that did *not* tip into Emergency that were forecast to be in Emergency.

5 Type I errors: false alarms

We define an emergency warning as a forecast, issued four to eight months ahead, that an area would be in Emergency or worse in a month for which FEWS NET later published a current-situation map. There were 4,084 such area-month warnings between 2011 and 2024.

Figure 5 shows what happened to them. In 46 percent of cases the area was in Emergency or worse, as forecast. In 54 percent it was not. Almost all of those ended one phase lower, in Crisis; only 3 percent ended in Stressed or better.

FEWS NET's aid flag lets us ask how many of these apparent false alarms FEWS NET itself attributes to aid. Of the warned areas that ended in Crisis, 60 percent carried the flag, against 12 percent of Crisis areas that had not been warned about. Taken at face value, the flag reclassifies 30 percent of all emergency warnings as crises held back by aid, leaving 20 percent ending in Crisis without the flag and 3 percent ending lower. On this reading, about a quarter of FEWS NET's emergency warnings were genuine false alarms.

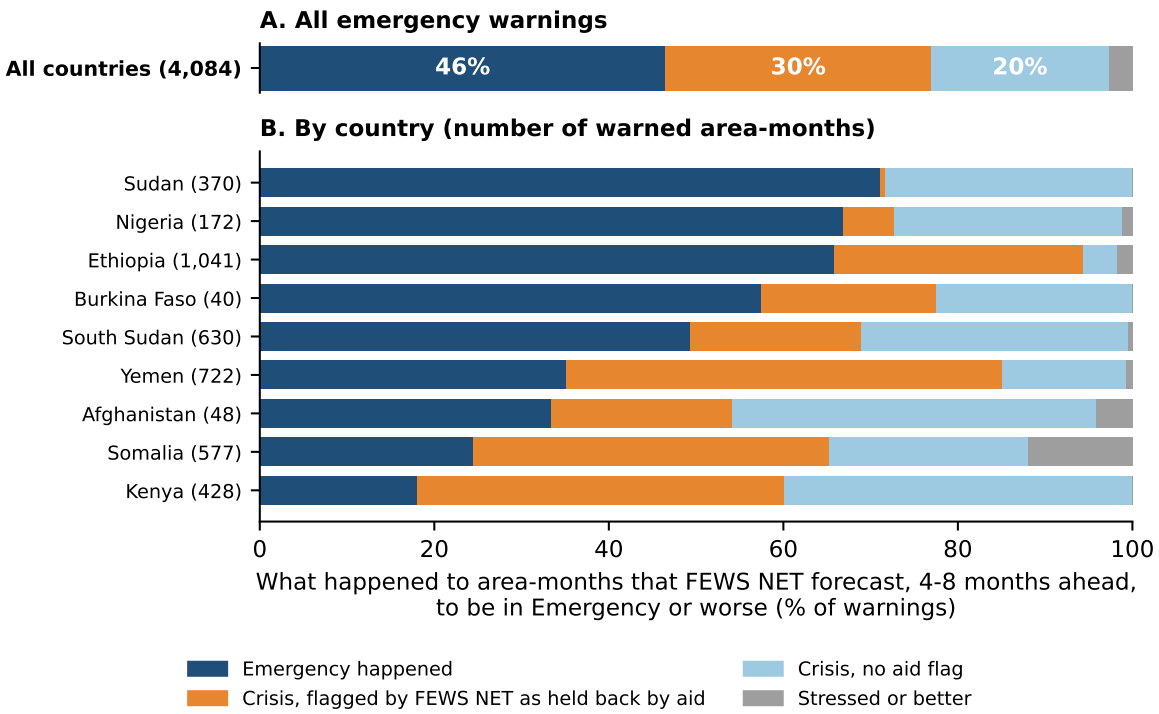


Figure 5: What happened to emergency warnings. Area-months that FEWS NET forecast, four to eight months ahead, to be in Emergency or worse (IPC Phase 4+), 2011–2024, by the phase FEWS NET later mapped. Panel A: all countries. Panel B: countries with at least 40 warned area-months. "Held back by aid" means FEWS NET's map carried its "!" flag: humanitarian assistance was keeping the area at least one phase better than it would otherwise be.

The pattern differs sharply across countries (Figure 5, panel B). In Sudan, Nigeria and Ethiopia, two-thirds or more of emergency warnings came true. In Somalia and Kenya, fewer than a quarter did, and FEWS NET credited aid with holding back about half or more of the rest. In Sudan, where warnings failed, FEWS NET almost never credited aid.

Two checks suggest caution before crediting aid with the rest.

First, 48 percent of the warned areas that ended in flagged Crisis had carried the flag on the map before the warning was issued. In those areas, the flag describes aid that was already flowing, not aid mobilized by the warning. The forecast should already have accounted for it, since FEWS NET's forecasts assume planned assistance. One reading is that FEWS NET forecast Emergency on the expectation that existing aid would fall away, and it did not; another is that the flag rationalizes a forecast that was too pessimistic. The data cannot distinguish them.

Second, if aid were preventing warned emergencies, warnings should come true less often when money then arrived. We test this in Appendix B using FEWS NET's "distinctive" warnings — the part of each forecast that rainfall, conflict, prices and the previous maps do not predict (Section 6). We compare countries whose humanitarian funding over the following six months rose more than their usual level and recent trend would predict with countries whose funding rose less. The share of distinctive warnings borne out is 0.50 where funding rose less than usual and 0.49 where it rose

more. Scoring against need (the mapped phase plus one if flagged) gives 0.51 and 0.49. We find no sign that funding surprises mask accuracy. Funding is only observed by country, so this test is coarse; it does not rule out well-targeted aid within countries.

Defining emergencies by need gives a similar picture. If we count an area as reaching Emergency when its mapped phase plus the aid flag reaches 4, there were 2,338 such cases, and 55 percent were held back by aid on FEWS NET's own account. But 78 percent of those held back were areas already in Crisis whose phase did not change; the flag simply switched on. And among these need-defined emergencies, early warning did not raise the share held back by aid (-6 percentage points, standard error 5). FEWS NET's forecasts already assume the aid that is coming, so they are not designed to answer this question.

5.1 Famine

Famine (Phase 5) is the outcome early warning exists to prevent, but it is too rare for error rates. Between 2011 and 2024 FEWS NET mapped famine in 98 of its units, in four episodes: Somalia in 2011, which accounts for almost all of them, South Sudan in 2017 and 2021, and Sudan in 2024. In none of them did its most likely forecast, made four to eight months ahead, put the area in famine. For Somalia in 2011 the earliest forecast the data warehouse holds was made three months ahead, and it placed the famine-affected areas in Crisis or Emergency; in South Sudan and Sudan the forecasts four months ahead were Emergency, and famine appeared in the most likely forecast only one or two months before. Conversely, FEWS NET forecast famine four to eight months ahead for only 6 unit-months that it later mapped: five in Somalia, in projections made in late 2022 for early 2023, and one in South Sudan in 2021. None was in famine when mapped, and FEWS NET credited aid with holding back 50 percent of them.

FEWS NET's written reports tell a different story from its maps. In the reports we extracted (Appendix F), FEWS NET raised the risk of famine in Somalia from January 2011, at least 6 months before famine was mapped, and in Sudan from June 2023, 12 months before. For famine, early warning has come as a statement of risk, often in a worst-case scenario, rather than as the most likely outcome on the map. That is a reasonable way to warn of a rare, catastrophic event, but it means the maps alone understate how early FEWS NET saw famine coming, and that scoring the maps cannot evaluate famine warnings with so few cases.

6 Does FEWS NET know something others don't?

Accuracy against FEWS NET's own later maps says little about value. What matters for a funder is what would be lost without FEWS NET: its information relative to what donors could otherwise see, scored against outcomes it did not produce. We approach this in two ways.

6.1 Against a world without FEWS NET

We define two benchmarks for what donors would know without FEWS NET. The first uses only raw signals: local rainfall, conflict and food prices before the forecast, plus country and season. The second adds the latest official IPC or Cadre Harmonisé analysis of the area, if one was made in the previous eight months. The second benchmark is generous to the world without FEWS NET, because FEWS NET staff contribute to many official analyses. It is also usually unavailable: only 25 percent of FEWS NET's area-rounds had a recent official analysis.

For each benchmark, we use gradient-boosted trees to predict two things out of sample (each year predicted from models fitted on the other years): FEWS NET's own current map and forecast, and the outcome in the next official analysis. "FEWS NET's contribution" is the part of its map or forecast that the benchmark does not predict. We then score everything against the next independent IPC or Cadre Harmonisé analysis of the area. We focus on areas that were not in Crisis at the latest official analysis, because predicting that a crisis continues is easy. This leaves 12,776 area-rounds in which 1,009 areas tipped into Crisis.

The main result is the area under the ROC curve (AUC): the probability that, of two areas, one that tipped into Crisis and one that did not, the model ranks the first higher (0.5 is a coin flip, 1 is perfect). Adding FEWS NET's map and forecast to the raw signals raises the AUC from 0.79 to 0.85, closing more than a quarter of the gap to perfect ranking. Adding them to the more generous benchmark raises it from 0.84 to 0.86.

The second gain is small, and it is worth being clear about why it does not undercut the rest of the paper. First, it is measured where FEWS NET's alternatives are strongest. The benchmark includes the latest official analysis, which exists for only 25 percent of FEWS NET's area-rounds, and every observation in this test is, by construction, an area the IPC or Cadre Harmonisé covers. Second, those official analyses are not independent of FEWS NET: its analysts take part in them and its data feed them, so the benchmark already contains much of what FEWS NET knows. Third, the AUC is close to its ceiling; the move from 0.84 to 0.86 closes about an eighth of the remaining gap, and the regressions below find the gain precisely estimated. The fair reading is that where a recent official analysis exists, FEWS NET adds modestly to it. FEWS NET's distinctive value lies mainly where none exists — three-quarters of its coverage — and there we cannot score it against an independent outcome. Section 7 shows that funding behaves accordingly.

Table 2 gives the regression version. The outcome is whether the area is in Crisis or worse at the next official analysis. We regress it on the benchmark's own prediction of that outcome and on FEWS NET's contribution, with country-by-round fixed effects, so the comparison is between areas in the same country at the same time:

$$y_{a,t+1} = \alpha\, \hat{y}_{at} + \beta\, c_{at} + \mu_{c(a),t} + \varepsilon_{at},$$

where $y_{a,t+1}$ is an indicator for Crisis or worse in area a at the next official analysis, $\hat{y}_{at}$ is the benchmark's prediction of it, c_{at} is FEWS NET's contribution (its map or forecast minus the

Table 2: Is FEWS NET's distinctive information borne out by independent assessments?

	Without FEWS NET: rainfall, conflict and prices			Without FEWS NET: rainfall, conflict, prices and latest official analysis		
	(1)	(2)	(3)	(4)	(5)	(6)
FEWS NET current map, beyond the baseline	0.201***		0.127***	0.165***		0.121***
	(0.015)		(0.023)	(0.013)		(0.020)
FEWS NET forecast, beyond the baseline		0.195***	0.131***		0.142***	0.082***
		(0.011)	(0.019)		(0.016)	(0.023)
Baseline's own prediction (coefficient)	0.42	0.42	0.44	0.68	0.68	0.68
Observations	12,764	12,764	12,764	12,764	12,764	12,764
Countries	21	21	21	21	21	21
AUC: baseline → adding FEWS NET		0.79 → 0.85			0.84 → 0.86	

Notes: Linear probability models. Outcome: area classified in Crisis or worse (IPC Phase 3+) at the next official IPC or Cadre Harmonisé analysis. Sample: official areas not in Crisis at the latest official analysis, 2011–2024. FEWS NET's map and forecast are overlaid onto official areas (share of the area in Crisis or worse). "Beyond the baseline" is FEWS NET's map or forecast minus its out-of-sample prediction from the baseline (gradient-boosted trees, fitted on other years). All columns control for the baseline's own out-of-sample prediction of the outcome and include country-by-round fixed effects. Standard errors clustered by country in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The last row gives the AUC for ranking areas that tip into Crisis, using the baseline alone and adding FEWS NET's map and forecast.

benchmark's prediction of that map or forecast), and $\mu_{c(a),t}$ is a country-by-round fixed effect. The coefficient β measures how much of FEWS NET's distinctive signal is borne out by an independent assessment. Over the raw-data benchmark, it is 0.13 for the current map and 0.13 for the forecast when both enter (column 3); over the benchmark with official analyses, 0.12 and 0.08 (column 6). All are precisely estimated.

The coefficients are well below one. Only part of FEWS NET's distinctive signal shows up in the next official assessment. Some of the gap is noise in FEWS NET's judgments, some is noise in the official assessments, and some, if aid works, is masking. But the signal is clearly there, and it survives controlling for the official analyses that do exist.

6.2 Head to head with the official projections

The IPC and Cadre Harmonisé also publish forecasts. Where both exist, we compare FEWS NET's forecast with the official projection for the same area and period, both scored against the next official current analysis. We match each official projection to FEWS NET's forecast from its latest round at or up to four months before the official analysis. This gives 6,683 area-projections in 14 countries; 4,128 of them were not in Crisis at the latest official analysis, and 227 of those tipped into Crisis. In the Cadre Harmonisé, projections target the June–August lean season, which is rarely assessed as a current period; we score them against the next current analysis (September–December). Both forecasters face the same mismatch.

Figure 6 plots both forecasts in Type I/Type II space. The vertical axis is the share of new crises that were not warned about (misses); the horizontal axis is the share of areas that stayed out of Crisis but were warned about (false alarms). Each forecast can be turned into a warning at any threshold — warn if at least 10 percent of the area is forecast in Crisis, or 50 percent, and so on — and each curve traces the misses and false alarms of one forecaster across all thresholds. The figure's main message is that the two forecasters are about equally skilled: at the false alarm

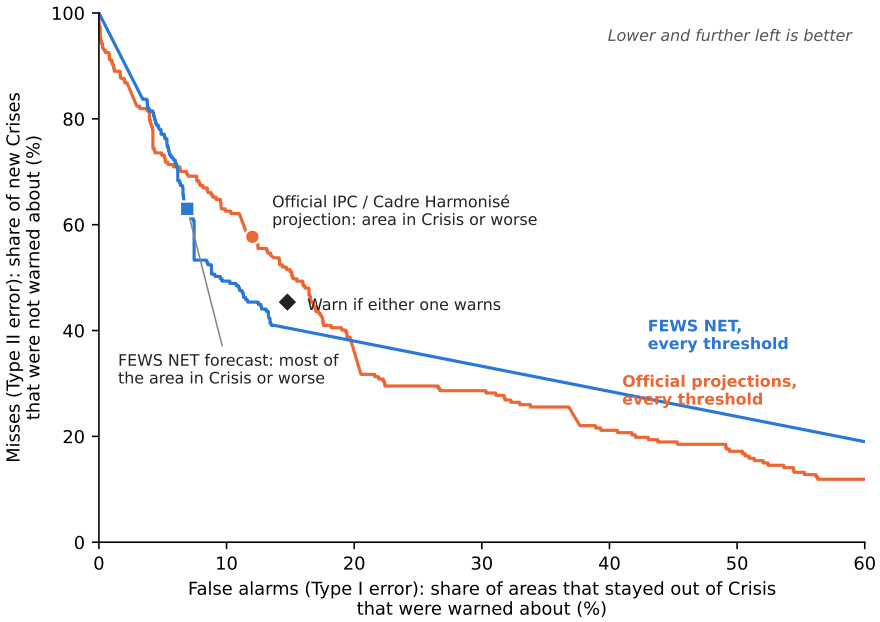


Figure 6: FEWS NET and the official projections in Type I/Type II space. 4,128 official IPC and Cadre Harmonisé areas not in Crisis at the latest official analysis, in 12 countries, 2014–2025; 227 tipped into Crisis at the next official analysis. Points: misses (new crises not warned about) and false alarms (areas warned about that stayed out of Crisis) for the official projection classifying the area in Crisis or worse, FEWS NET forecasting Crisis or worse for most of the area, and either one warning. Lines trace every warning threshold for the official projection (orange: share of the population projected in Crisis or worse) and for FEWS NET (blue: share of the area forecast in Crisis or worse); FEWS NET forecast none of the area in Crisis for most of these areas, so beyond that point its curve is a straight line. Lower and further left is better. The official projections are made with FEWS NET participation and after FEWS NET's matched forecast.

rates either one actually operates at, below about 20 percent, FEWS NET's curve lies on or slightly below the official one, and the two cross only where both would be raising false alarms for a fifth or more of quiet areas. The official projection classifies the area in Crisis for 42 percent of new crises, at a false alarm rate of 12 percent. FEWS NET forecasts Crisis for most of the area in 37 percent of new crises, at a false alarm rate of 7 percent. FEWS NET is more conservative: it misses slightly more but cries wolf less. The two are complements. Warning when either one warns catches 55 percent of new crises at a false alarm rate of 15 percent. In 12 percent of new crises, FEWS NET warned and the official projection did not.

Table 3 regresses the next official classification on both forecasts. FEWS NET's forecast predicts the outcome over and above the official projection. With both forecasts and the latest official analysis (column 4), the coefficient on FEWS NET's forecast is 0.18 and on the official projection 0.06. For areas not yet in Crisis (Panel B, column 3), they are 0.15 and 0.05. With only 12 to 14 countries, cluster-robust standard errors can overstate precision; wild cluster bootstrap p-values for FEWS NET's coefficient are 0.003 and 0.006.

Neither forecaster is independent of the other. FEWS NET analysts join many official analyses, FEWS NET's maps and data feed them, and each team reads the other's work, so the official

Table 3: FEWS NET's forecasts and the official projections, scored against the next official analysis

	(1)	(2)	(3)	(4)	(5)
A. All areas					
Official projection: Crisis or worse	0.220***		0.137***	0.056***	0.054***
	(0.057)		(0.030)	(0.018)	(0.017)
FEWS NET forecast: share of area in Crisis or worse		0.329***	0.239***	0.178***	0.106***
		(0.063)	(0.048)	(0.046)	(0.037)
Latest official analysis: Crisis or worse				0.275***	0.261***
				(0.027)	(0.024)
FEWS NET current map: share of area in Crisis or worse					0.127**
					(0.056)
Observations	6,683	6,683	6,683	6,559	6,559
Countries	14	14	14	14	14
B. Areas not in Crisis at the latest official analysis					
Official projection: Crisis or worse	0.104***		0.054***	0.054***	0.056***
	(0.015)		(0.014)	(0.014)	(0.014)
FEWS NET forecast: share of area in Crisis or worse		0.179***	0.149***	0.149***	0.104**
		(0.035)	(0.033)	(0.033)	(0.041)
FEWS NET current map: share of area in Crisis or worse					0.094
					(0.110)
Observations	4,128	4,128	4,128	4,128	4,128
Countries	12	12	12	12	12

Notes: Linear probability models. Outcome: area classified in Crisis or worse (IPC Phase 3+) at the next official IPC or Cadre Harmonisé current analysis after the projection. Official projection: Cadre Harmonisé March projections for June–August, and IPC first projections. FEWS NET forecast: share of the official area that FEWS NET's latest forecast (issued up to four months before the official analysis) placed in Crisis or worse for the projection period. All columns include country-by-analysis-round fixed effects. In Panel B, the latest official analysis is constant by construction, so column (4) repeats column (3). Standard errors clustered by country in parentheses; with 12–14 clusters, see wild bootstrap p-values in the text. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

projections partly embody FEWS NET's information and the comparison understates what FEWS NET adds. The timing tilts the same way. We compare each official projection with FEWS NET's forecast from a round up to four months earlier, so the official analysts could have seen that forecast, while FEWS NET could have seen only the previous official analysis. That makes it more notable, not less, that FEWS NET's forecast keeps predictive power after controlling for the official projection.

6.3 Where the information lies

FEWS NET bundles field analysts who draw maps and write reports with market price monitoring. If a funder could buy the information more cheaply in pieces, the pieces should carry it. They do not. FEWS NET's own price series, from 881 markets, add almost nothing to forecasts of new crises or escalation, whether on top of public data (AUC 0.725 without prices and 0.727 with them, against 0.901 with FEWS NET's map and forecast) or on top of FEWS NET's maps. FEWS NET's written reports, which we turned into data for Somalia, Ethiopia and Sudan with a large language model (Appendix F), add nothing measurable to its maps in predicting escalation from Crisis to Emergency (AUC 0.60 with the map alone, 0.59 with the text): the same analysts write both, and the maps already carry what the reports say. What public data cannot reproduce is the analysts' judgment, and that is the part independent assessments bear out.

7 Does money follow the warnings?

Information has value only if someone acts on it. We cannot observe aid at the level of FEWS NET's areas, so we ask whether humanitarian funding to a country moves with FEWS NET's classifications of that country. For each country and FEWS NET round, we regress the log of humanitarian funding committed over the next six months on the share of the country's FEWS NET areas in Crisis or worse on the current map, and on the share forecast to be in Crisis or worse four to eight months ahead.

Across countries, funding follows FEWS NET's maps closely. Without controls, and holding the current map fixed, a 10 percentage point rise in the share of a country's areas forecast to be in Emergency goes with 76 percent more humanitarian funding (95 percent interval 44 to 115). Most of this reflects persistent differences between countries: large, chronic crises get both bad maps and more money. Figure 7A shows what survives as controls are added for global funding swings, the country's funding in the previous six months, and the country's usual aid level. With all controls, the current map no longer predicts funding (−2 percent for food and nutrition funding per 10 percentage points of areas in Crisis), but the forecast does (25 percent; interval 5 to 48). Within a country, money moves when FEWS NET expects things to get worse, not when it reports that they are bad. The Emergency shares are too rarely nonzero within countries to estimate this precisely (Appendix Table A1).

Does funding respond to FEWS NET's own information, or to the rainfall failures and conflict that donors could see directly? Panel B splits the forecast into the part predictable from the country, the season and raw signals, and FEWS NET's contribution. The predictable part is mostly country and season: those alone account for 61 percent of its variation, and within a country and season, rainfall, conflict and prices explain only 5 percent of the variation in FEWS NET's forecasts. Across country-rounds, the predictable part varies little within countries (standard deviation 0.10, against 0.22 overall), while FEWS NET's contribution varies almost entirely within them (0.15 of 0.15). Only the contribution predicts funding: 12 percent more humanitarian funding (interval 4 to 21) and 33 percent more food and nutrition funding (11 to 59) per 10 percentage points of areas FEWS NET forecasts in Crisis beyond what that information predicts. The predictable part has no detectable effect (3 and 10 percent, with wide intervals). Results are similar against the benchmark that includes official analyses (Table A1). Funding follows FEWS NET's distinctive information more strongly where official analyses are scarce. In country-rounds where fewer than half of FEWS NET's areas had a recent official analysis (32 percent of country-rounds have more), a 10 percentage point rise in its distinctive forecast comes with 13 percent more humanitarian funding and 39 percent more food funding. Where official analyses are common, the figures are 8 and 16 percent. The difference is imprecise (standard errors 0.59 and 1.31 log points per unit), but it matches the pattern in Section 6.

These are associations, not causal effects. Donors and FEWS NET may respond to the same news that our raw signals miss — a displacement wave, a government request, a news story. But the pattern is the one we would expect if FEWS NET's forecasts inform allocation: funding responds

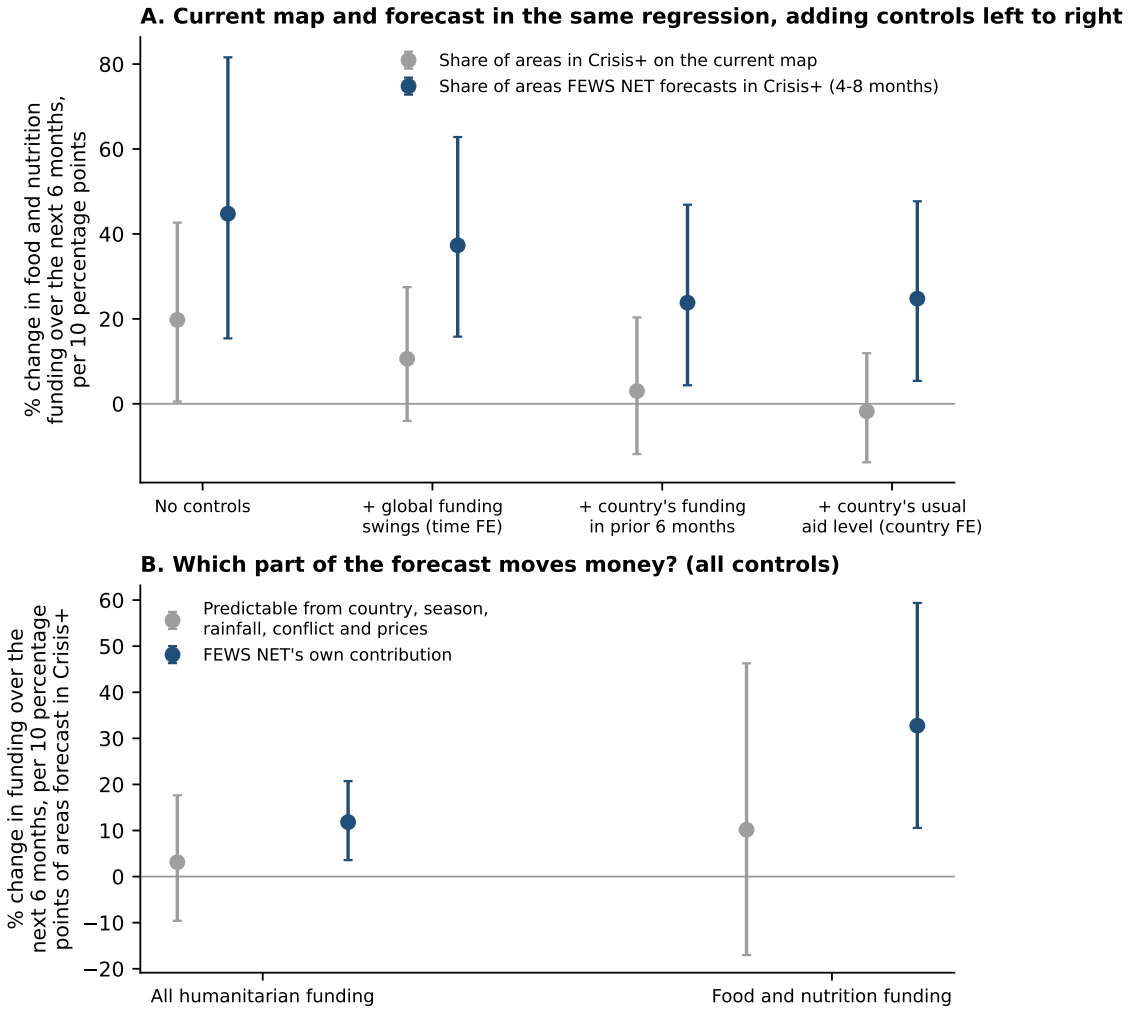


Figure 7: Does humanitarian funding follow FEWS NET? Country × FEWS NET round, 2011–2024. Outcome: humanitarian funding committed to the country over the next six months (OCHA Financial Tracking Service, log), expressed as the percent change per 10 percentage points more of the country's FEWS NET areas in Crisis or worse. Panel A: the share in Crisis or worse on the current map and the share forecast in Crisis or worse four to eight months ahead, in the same regression, adding controls from left to right. Panel B: the forecast split into the part predictable from rainfall, conflict, prices, country and season, and FEWS NET's own contribution; with time and country fixed effects and funding in the previous six months. 95 percent intervals clustered by country.

to the forward-looking, distinctive part of FEWS NET's analysis, and not to what anyone could infer from a country's usual situation and public data.

Within countries. Funding data are recorded by country, so we cannot see whether aid goes where FEWS NET points within a country. A partial check uses the small share of aid that can be located by district: the UN's country-based pooled funds, about 3–5 percent of humanitarian funding, together with records of which organizations work where and how many people they reached, for Somalia and Sudan (Appendix I). In Somalia the located money follows FEWS NET's forecasts, but only the part that public information predicts, and FEWS NET's next forecast predicts it about as well, so money and forecasts track the same conditions. In Sudan it does not follow the forecasts. FEWS NET's aid flags track where aid organizations work only loosely. Because located aid is such a small and atypical slice of the total, we treat these results as suggestive at most.

8 What is FEWS NET worth?

The question before the State Department is whether FEWS NET's information improves the use of food aid by enough to justify its cost. We answer in the unit FEWS NET is designed to forecast: person-months of food insecurity, here people kept out of Crisis or worse (IPC Phase 3+) for a month. FEWS NET can add value on two margins. With the food-aid budget fixed, its forecasts can help place the aid where people would otherwise go hungry; this is its potential value, if decision makers use it well. And its warnings can enlarge the budget itself; this is its historical influence, which depends on donors as much as on FEWS NET. We estimate both.

Why food insecurity, not lives. FEWS NET's phases, like the IPC's, classify acute food insecurity: whether households can eat enough, and what they give up to do so. That is what they track, if loosely. In WFP household surveys across 29 countries, the share of households with poor or borderline food consumption rises from 28 percent in areas FEWS NET maps as Minimal to 47 percent in areas in Crisis. Phases are a much weaker guide to child malnutrition and mortality: in Somalia's scheduled nutrition surveys FEWS NET's forecasts do not predict measured acute malnutrition (correlation 0.10), and across surveys crude death rates barely differ by phase. We document these patterns in a companion paper. Here we value FEWS NET in the unit it is designed to forecast and treat lives as a heavily qualified extension (Appendix H).

Cost. Federal contract records show FEWS NET's cost directly (USAspending.gov, 2026). Obligations under FEWS NET's contracts averaged $37 million a year over fiscal years 2017–2024 and $44 million over 2019–2023. They are lumpy, ranging from $22 million to a peak of $63 million in fiscal 2023, when several new task orders were funded up front; that peak is the source of a widely reported figure of $64 million a year (NPR, 2025). About two-thirds of the money goes to field analysis — the maps and forecasts — through the main task order ($139 million over 2019–2024).

The rest pays for the data hub, livelihood baseline studies, strategic planning and technical support at the U.S. Geological Survey's satellite data center. Interagency agreements with U.S. science agencies are not in the contract records and add a few million dollars more. We use $45 million a year as the central figure and $35–65 million as a range.

8.1 Placing a fixed budget

We model a planner who, in each country and FEWS NET round, has a fixed number of rations, each covering one person's food needs for a month, and places them across the country's areas to keep as many people as possible out of Crisis. By the IPC's definition, a person in Crisis has a food consumption gap; a ration closes it for a month, while a ration given to someone who would not have been in Crisis averts nothing. The planner predicts each area's share of the population in Crisis or worse at the next official IPC or Cadre Harmonisé analysis, which is independent of FEWS NET's own maps, using either public information (rainfall, conflict, prices, country, season and the latest official analysis, where one exists) or the same plus FEWS NET's current map and forecast. Predictions are made out of sample, by year. We compare these with planners who know only population, only the latest official analysis, or the outcome itself.

Two further choices matter. First, the size of the budget: we set it at 32 percent of the Crisis population's needs, the share that actual food security and nutrition funding to these countries would cover at the cost of a full ration, about $22.50 per person a month (29–36 percent for rations costing $20–25; World Food Programme, 2023; Somalia Cash Working Group and Food Security Cluster, 2022). Second, how well aid is targeted *within* an area once it arrives. If at random, a ration reaches someone in Crisis with probability equal to the area's Crisis share; if perfectly, it reaches someone in Crisis whenever there is one left. Studies of targeting in poor and crisis settings find large errors: community-based and proxy-means targeting typically leave a third to a half of transfers with households that are not among the intended poor (Sabates-Wheeler et al., 2015; Brown et al., 2018; Premand and Schnitzer, 2021), so reality lies between random and halfway to perfect. The sample covers 1,803 official areas in 24 countries, 2013–2024.

FEWS NET's forecasts help, modestly (Figure 8). With random targeting within areas, the share of rations reaching people who would otherwise be in Crisis rises from 32 percent with public information to 35 percent with FEWS NET's forecasts added; with targeting halfway to perfect, from 56 to 59 percent. That makes the same budget 4–8 percent more effective at reaching people in Crisis, and closes 17–20 percent of the gap to perfect foresight. FEWS NET's forecasts also beat the latest official analysis used alone. Targeting within areas matters more than any information: moving from random to perfect targeting raises effectiveness far more than FEWS NET does, and with perfect targeting FEWS NET's forecasts add little.

In the sample, FEWS NET's forecasts keep 2.4–2.6 million more person-months out of Crisis each year. That is 0.7–0.8 percent of the 346 million person-months of Crisis or worse in these areas each year: small against the scale of need, but large against FEWS NET's cost. At $22.50 a ration, it is the equivalent of $54–59 million a year of additional food aid, against FEWS NET's

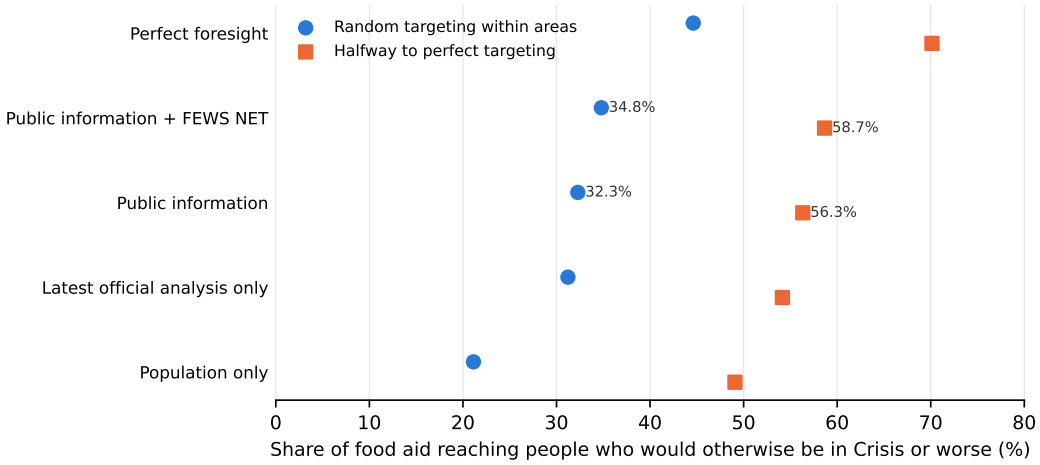


Figure 8: Placing a fixed food-aid budget: the share of rations reaching people who would otherwise be in Crisis or worse, by what the planner knows. Budget: 32 percent of the Crisis population's needs in each country and FEWS NET round. Circles: rations distributed at random within areas; squares: halfway between random and perfect targeting within areas. Outcome: the share of each area's population in Phase 3 or worse at the next official IPC or Cadre Harmonisé analysis. 1,803 areas in 24 countries, 2013–2024. Predictions out of sample, by year.

budget of $45 million: a ratio of 1.2–1.3.

Extrapolating to all of FEWS NET's countries. The sample covers only areas with a following official analysis, mostly from 2017 outside West Africa; FEWS NET covers 37 countries. Food security and nutrition funding to all FEWS NET countries averaged $4.0 billion a year over the same years, against $2.5 billion to the country-years in the sample. If FEWS NET's forecasts improve the placement of each aid dollar elsewhere as much as in the sample, scaling by funding (a factor of 1.6) gives 3.9–4.2 million person-months a year, worth $88–95 million of food aid: 2.0–2.1 times FEWS NET's cost (1.2–3.0 across ration costs and FEWS NET's budget range). This extrapolation is an assumption, not an estimate. It could be conservative, since FEWS NET's information is most distinctive where official analyses are missing (Section 6), or generous, if aid in those countries is harder to place well.

8.2 Enlarging the envelope

The second margin is how much aid FEWS NET's warnings brought in. Within countries, a 10 percentage point rise in the share of a country's areas that FEWS NET forecast to be in Crisis, beyond what public information predicted, came with about 28 percent more food security and nutrition funding over the next six months (Section 7). Applying that response to FEWS NET's distinctive forecasts country by country, and counting the funding it subtracted where it was calmer than public information as well as the funding it added, a net 7–9 percent of food security and nutrition funding to FEWS NET's countries followed its distinctive information (16 percent counting only the additions). That is $274–370 million a year. If that money was additional

and was placed as in the planner's problem above, it kept 4.2–9.6 million person-months a year out of Crisis. If instead it was drawn from other crises where it would have done as much good, the net gain is close to zero. And the funding response is an association: donors and FEWS NET may both respond to news our public data miss. We therefore treat the envelope as an upper bound on historical influence, not as an estimate of what FEWS NET caused.

8.3 Putting it together

For the State Department, the two margins answer different questions. The reallocation margin asks what FEWS NET's information is worth if the government and its partners use it to place a given food-aid budget: on our estimates, around one to two times its cost in additional food aid, depending on the extrapolation, or about $11–11 of FEWS NET's budget per person-month kept out of Crisis across all its countries, against $22.50 for the ration itself. The envelope margin asks what FEWS NET's warnings have done to the size of the budget: historically, a substantial share of food aid moved with them, worth several times FEWS NET's cost if the money was new. Neither margin requires that FEWS NET's forecasts be perfect, only that they be better than what decision makers would otherwise know, which Sections 6 and 7 show they are.

Three cautions apply. The reallocation value is potential: the within-country evidence for Somalia and Sudan does not show located aid responding to FEWS NET's distinctive information (Appendix I). The extrapolation assumes the sample is representative. And the calculation counts only people kept out of Crisis; it gives no extra weight to keeping people out of Emergency or Famine, where the stakes are highest.

And lives? Converting person-months of food insecurity into deaths requires the link between phases and mortality, which is weak (Appendix G). Appendix H carries out the conversion anyway, taking FEWS NET's own judgments of where aid holds back a crisis at face value and using the evidence-based calibration of deaths per phase. It puts the cost of FEWS NET's distinctive information at $46,000–150,000 per death averted, or $3,700–15,000 if the IPC's reference bands are taken literally. We do not think deaths are the right yardstick for a system built to measure food insecurity, and we report the conversion only to show how much it depends on that link.

9 What the data cannot tell us

The ideal evaluation would compare welfare — child malnutrition, mortality — in places and times with and without FEWS NET. We tried two versions with the data available, and neither is informative.

The 2025 shutdown. FEWS NET went dark in January 2025 just as U.S. aid was cut. If its "held back by aid" flags are right, flagged areas should have deteriorated faster afterwards than comparable unflagged areas. Over 12 months, flagged areas worsen relative to comparable unflagged

areas by about 0.20 of a phase in normal years. For windows ending in 2025, the gap widened by a further 0.09 phases on FEWS NET's own later maps (wild bootstrap $p = 0.43$) and 0.15 on official classifications ($p = 0.26$). The flag carries information about fragility, but the data cannot yet separate the cut's effect from the trend. We need outcomes beyond mid-2026 (Appendix C).

Changes in FEWS NET coverage. FEWS NET has entered and left countries over time, but these moves follow aid and conflict trends, so comparing levels before and after is confounded. A narrower test asks whether local shocks do less damage, within a country, when FEWS NET covers it. Local rainfall, conflict and price shocks explain only 2.0 percent of the variation in official classifications within country and month, and coverage does not change their effect (-0.010, standard error 0.019, before 2025). Classifications move too slowly, and too nationally, to detect FEWS NET's effect this way (Appendix D). The same weakness of public fundamentals appears in Section 7: they explain little of FEWS NET's forecasts within a country and season, which is why most of FEWS NET's information cannot be reproduced from public data.

Testing whether FEWS NET saves lives will require outcomes it does not produce: child anthropometry and mortality from household surveys, linked to the timing and location of its warnings.

10 Conclusion

FEWS NET rarely misses a food crisis outright, but it often sees emergencies late and one phase too mild. About half of its emergency warnings do not come true; FEWS NET credits aid for about half of those, though often aid that was already flowing. Measured against what donors could otherwise know, it adds information that independent assessments later bear out, it raises fewer false alarms than the official projections, and humanitarian funding moves with the part of its forecasts that public data cannot reproduce.

For the State Department, the relevant comparison is with what would replace FEWS NET. In three-quarters of FEWS NET's coverage, there is no recent official analysis to fall back on, and the raw signals — rainfall, conflict, prices — predict where crises will strike much less well. FEWS NET's distinctive contribution is modest in any single forecast but systematic. Used to place a fixed food-aid budget, it would keep roughly 3.9–4.2 million more person-months a year out of Crisis across FEWS NET's countries, worth about 2.0–2.1 times its cost in food aid, on an extrapolation we flag as uncertain. Historically, a substantial share of food aid moved with its warnings, though some of that money would have gone to other crises anyway. On both margins, FEWS NET's $45 million a year — around one percent of the food aid it informs — looks well spent, provided the people who allocate aid use what it tells them. Measured in lives, the case is weaker and much less certain, because a phase of food insecurity says little about mortality; that is a limit of what FEWS NET measures, not of how well it measures it.

For funders choosing among pieces rather than the whole, the value lies in the analysts' judgment. Neither FEWS NET's written reports nor market price series add much to what its maps

already say. What is missing is outcome measurement: without repeated nutrition and mortality surveys in areas at Crisis or worse, there is no way to tell afterward how bad a crisis was or whether the response worked, and FEWS NET's phases, which classify food insecurity, cannot stand in for them.

Three changes would make the system easier to evaluate and probably more useful. First, publish a forecast without assumed aid alongside the current one. Donors need to know what will happen if they do nothing, and evaluators need it to separate crises prevented from false alarms. Second, record when and why the aid flag is applied, so that "held back by aid" can be distinguished from "aid was already there." Third, validate forecasts against independent outcomes. Our comparisons use the IPC and Cadre Harmonisé, which share analysts with FEWS NET; publishing the household indicators behind each classification, as the Cadre Harmonisé has begun to do, would let anyone check them.

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A Data construction

FEWS NET classifications. We download all current-situation ("CS") and projection ("ML1" near-term, "ML2" medium-term) classifications from the FEWS NET Data Warehouse, month by month, January 2011 to August 2026, and use those through December 2024. Each record gives an area identifier, its phase, the projection period and whether the area carries the aid flag. We use the full map rounds (three per year since 2016; quarterly before). For the analyses in Sections 6 and 7, the unit is an area and map round, giving 190,131 area-rounds for 16,053 areas in 28 countries with complete forecasts and a next map within eight months.

Official classifications. Cadre Harmonisé records (current and projected) come from the regional dataset; IPC records from the IPC's area-level data (2017–2026). For IPC records we derive the area phase from population shares using the IPC's 20 percent rule. Areas are matched to polygons from the Cadre Harmonisé 2023 map, IPC country maps and geoBoundaries administrative units, and overlaid on FEWS NET's areas in an equal-area projection. FEWS NET's classifications are averaged over each official area, weighted by overlap.

Raw signals. Rainfall is the area mean of monthly CHIRPS rainfall. Conflict is ACLED's district-level political violence data, available sub-nationally for the 19 countries with UN humanitarian response plans. Districts are matched to OCHA administrative boundaries by district code, then by name, and apportioned to FEWS NET areas by area share. Prices are the mean 3- and 12-month log changes of retail cereal and tuber prices at the WFP market nearest each FEWS NET area, within 150 km.

Funding. Humanitarian commitments ("paid" and "commitment" status, incoming flows) from the Financial Tracking Service, assigned to recipient countries; multi-country flows are dropped. Food and nutrition funding is the subset whose cluster is Food Security or Nutrition.

Prediction models. All benchmark predictions use gradient-boosted trees (scikit-learn's histogram gradient boosting) with country as a categorical feature, cross-fitted by year: predictions for each year come from a model trained on all other years.

B The masking test

If aid prevents crises that FEWS NET warns about, its distinctive warnings should be borne out less often when funding to the country rises unexpectedly. Funding surprises are country-round residuals of log funding over the next six months, after removing country and time effects and the country's funding in the previous six months. We split country-rounds into thirds and estimate, within each third, how much of FEWS NET's distinctive warning (the forecast minus its prediction from raw signals and previous maps) is borne out on its next map, with country-by-round fixed

effects. Figure A1 shows no pattern: the share is about one half in all three groups, whether outcomes are scored as mapped or by need. The interaction of the distinctive warning with the funding surprise is 0.009 (standard error 0.028).

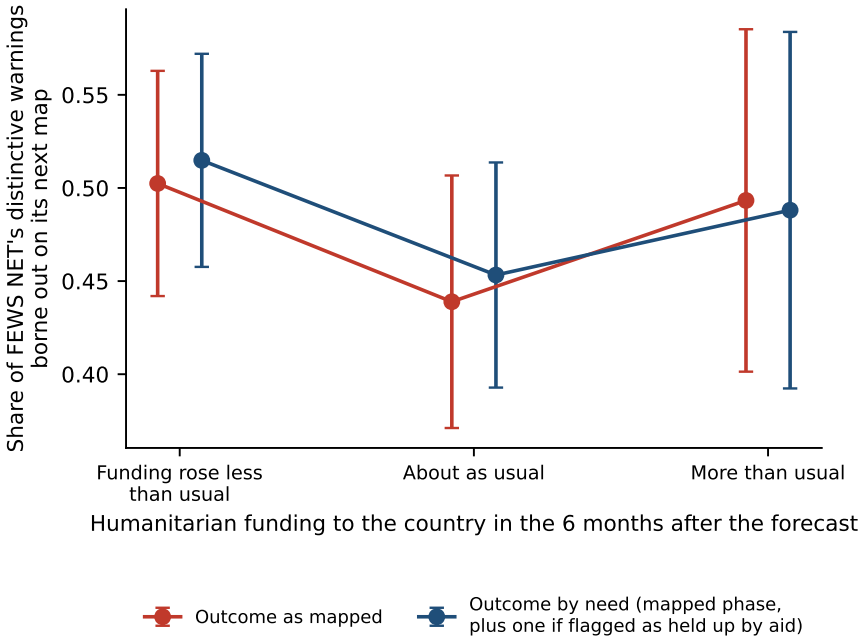


Figure A1: Do warnings come true less often where money flowed? Coefficient on FEWS NET's distinctive warning (forecast of Crisis or worse minus its out-of-sample prediction from rainfall, conflict, prices, previous maps, country and season) in a regression of Crisis or worse on its next map, by the country's funding surprise over the following six months. Red: outcome as mapped. Blue: outcome by need (mapped phase plus one if flagged as held up by aid). 95 percent intervals clustered by country.

C The 2025 shutdown

We compare same-season changes in areas FEWS NET flagged as held up by aid with unflagged areas that started in the same phase, country and month, for 12-month windows ending each year from 2014 to 2025. Windows ending in 2025 span the shutdown and the U.S. aid cut; earlier windows show how flagged areas normally evolve. Figure A2 shows that flagged areas worsen relative to unflagged ones in almost every year, and that the gap was already rising in 2023–2024, when humanitarian funding was falling. The 2025 gap is in line with that trend. Using 24-month windows produces a larger 2025 gap that is not repeated in 2026; pooling horizons, as in an earlier version of this analysis, mixes windows of different lengths in different years and overstates the effect.

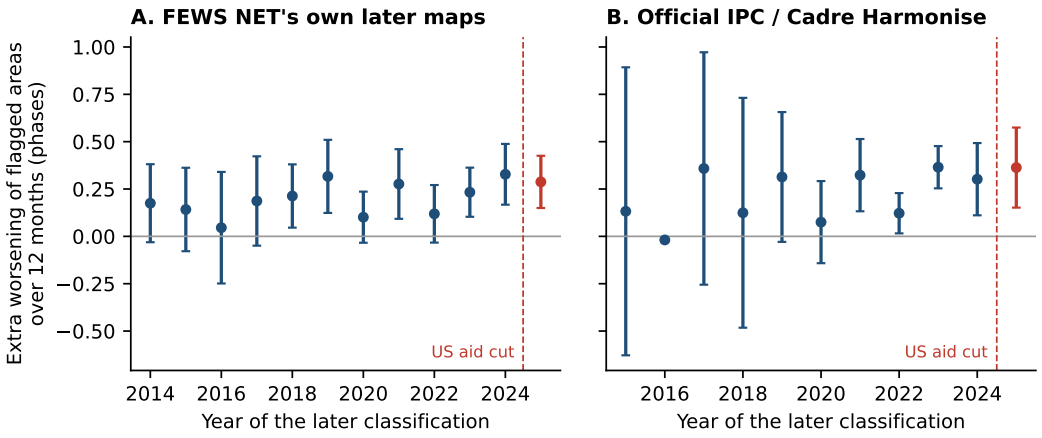


Figure A2: Did areas FEWS NET flagged as held up by aid deteriorate after the 2025 shutdown? Difference in the 12-month change in phase between flagged and unflagged areas that started in the same phase, country and month, by the year the window ends. Panel A: FEWS NET's own maps (FEWS NET areas in Phase 2–3). Panel B: official IPC and Cadre Harmonisé analyses, with the flag measured as the share of the official area covered by flagged FEWS NET areas in a map made at most four months before. Red: windows ending after the shutdown. 95 percent intervals clustered by country.

D Coverage changes and local shocks

We built an index of local shocks — rainfall deficits, political violence and food price spikes, weighted to predict official classifications in other countries — and asked whether its effect on official classifications differs, within a country, between periods when FEWS NET covered the country and periods when it did not. With area and country-by-month fixed effects and country-specific shock effects, the interaction is −0.010 (standard error 0.019) before 2025 and 0.002 (0.006) in 2025–2026, measured on the share of the population in Crisis or worse. Political violence is apportioned to each official area from ACLED's districts, placed on OCHA boundaries by district code, by the share of each district inside the area, as for FEWS NET's areas (Appendix A). On the indicator for an area being in Crisis or worse, rather than the population share, the pre-2025 interaction is positive (0.130, standard error 0.054): if anything, shocks moved classifications more under FEWS NET coverage, which may reflect classifications responding faster where FEWS NET is present rather than worse outcomes. The index explains only 2.0 percent of the within-country variation in classifications; rainfall alone explains 0.06 percent. An earlier specification suggested that shocks did *more* damage under FEWS NET coverage; that result rested entirely on Cameroon, where FEWS NET arrived in 2020 as conflict escalated, and on archive gaps that made FEWS NET appear to leave Burkina Faso and Mauritania when it had not.

E Funding regressions

Table A1 reports the funding regressions behind Figure 7 and the funding responses used in Section 8.

Table A1: Humanitarian funding and FEWS NET's maps and forecasts, within countries

	Map vs forecast		Beyond rainfall, conflict and prices		Beyond rainfall, conflict, prices and official analysis	
	All (1)	Food (2)	All (3)	Food (4)	All (5)	Food (6)
Share of areas in Crisis+, current map	−1.0	−1.8				
	(4.0)	(6.7)				
Share of areas forecast in Crisis+	10.0***	22.1**				
	(3.8)	(8.6)				
Forecast: part predictable from country, season and shocks			3.1	9.7	6.7	16.3
			(6.7)	(14.5)	(6.2)	(12.8)
Forecast: FEWS NET's own contribution			11.2***	28.3***	10.2***	26.9***
			(3.9)	(9.3)	(3.6)	(9.4)
Country × map rounds	968	967	851	850	851	850
Countries	37	37	28	28	28	28

Notes: Country × FEWS NET map round, 2011–2024. Outcome: log humanitarian funding committed to the country over the six months after the map (OCHA Financial Tracking Service); all humanitarian funding ("All") or food security and nutrition ("Food"). Coefficients are scaled to log points per 10 percentage points of the country's FEWS NET areas (approximately the percent change). Columns 1–2: share of areas in Crisis or worse on the current map and in the medium-term forecast, in the same regression. Columns 3–6: the forecast share split into its out-of-sample prediction from rainfall, conflict, prices, country and season (columns 3–4), or these plus the latest official analysis (columns 5–6), and FEWS NET's contribution beyond that prediction. All columns include country and time fixed effects and log funding in the previous six months. Standard errors clustered by country in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F FEWS NET's reports and escalation

FEWS NET publishes far more than maps. Each outlook sets out the assumptions behind the forecast, a worst case, the aid expected and the data the analysts relied on, and updates, key messages and alerts follow between rounds. If the maps compress this reasoning too far, the text might hold early signs of the step FEWS NET forecasts worst, from Crisis to Emergency (Section 4). The same records give the famine-risk statements used in Section 5.1.

We tested this for Somalia, Ethiopia and Sudan, 2011–2024. We downloaded all 502 of FEWS NET's reports for these countries, including 135 full outlooks, and used a large language model (Claude) to turn each into structured records. For each area discussed, we recorded the phases stated now and for the projection, any worst case, the expected direction, the drivers named, the status of aid and any statement that data were missing. We also recorded every survey result the report quotes. The model was told to record only what each report says. This matters because the model knows how these crises turned out. Recording stated facts is safe, but asking the model to forecast past crises would not be. Each area is matched to FEWS NET's mapping units by name. As a check, where a report names a district or livelihood zone, the current phase it states lies within the phases on FEWS NET's map for those units in 91 percent of 322 cases.

The text adds little to the maps. Of 8,564 area-rounds in Crisis, 744 reached Emergency on the next map; FEWS NET had forecast Emergency for 32 percent of them. We add everything extracted from the reports to FEWS NET's forecast and aid flags: projected Emergency, worst cases, expected deterioration, expected cuts in aid, access constraints, the drivers named and statements of uncertainty. The cross-fitted AUC for escalation does not rise (0.60 with the map alone, 0.59 with the text). Scored by need, it barely moves (0.71 to 0.73). Within country and

round, two features of the text are associated with escalation (Table A2). Areas where a report expected aid to fall, or said access was constrained, were 6 and 7 percentage points more likely to reach Emergency, against a base of 8.7 percent. Both concern the assumption that planned aid will arrive, which FEWS NET's forecasts build in. But neither improves out-of-sample prediction by more than a hundredth of AUC. The same analysts write the reports and draw the maps, and the two mostly say the same thing. What FEWS NET knows is largely already in its maps. Mining its reports is not a route to much better forecasts, though where its reports doubt the aid pipeline, its forecasts deserve less confidence.

Table A2: Do FEWS NET's reports predict escalation from Crisis to Emergency beyond its maps?

	Emergency+ on next map			Emergency+ by need		
	(1)	(2)	(3)	(4)	(5)	(6)
FEWS NET forecast: Emergency	0.245***	0.242***	0.245***	0.408***	0.409***	0.406***
	(0.040)	(0.037)	(0.036)	(0.062)	(0.062)	(0.060)
Aid flag now	0.033	0.033	0.035	0.253***	0.250***	0.242***
	(0.033)	(0.036)	(0.036)	(0.036)	(0.037)	(0.036)
Aid flag in forecast	0.117***	0.113***	0.114***	0.211***	0.207***	0.197***
	(0.019)	(0.018)	(0.019)	(0.044)	(0.046)	(0.047)
Report: Emergency projected		0.002	-0.001		0.003	-0.001
		(0.030)	(0.031)		(0.033)	(0.031)
Report: worst case Emergency or Famine		0.000	-0.001		-0.002	-0.007
		(0.021)	(0.020)		(0.029)	(0.030)
Report: deterioration expected		-0.025	-0.026		-0.039	-0.035
		(0.019)	(0.019)		(0.031)	(0.032)
Report: aid expected to fall		0.060**	0.050*		0.063**	0.050*
		(0.030)	(0.027)		(0.029)	(0.026)
Report: access constrained		0.067***	0.059***		0.037	0.031
		(0.024)	(0.022)		(0.035)	(0.030)
Report: conflict or displacement		-0.015	-0.010		0.006	0.010
		(0.016)	(0.016)		(0.019)	(0.021)
Report: poor rains or harvest		-0.003	0.004		-0.005	0.005
		(0.034)	(0.031)		(0.032)	(0.032)
Report: high prices or currency		0.021	0.020		0.003	0.004
		(0.020)	(0.020)		(0.026)	(0.027)
Report: uncertainty stated		-0.008	-0.015		0.027	0.014
		(0.026)	(0.025)		(0.050)	(0.053)
Area discussed in reports		-0.059	-0.071		-0.081*	-0.098**
		(0.050)	(0.049)		(0.048)	(0.045)
Survey in previous 12 months			0.091***			0.127***
			(0.029)			(0.028)
Survey: acute malnutrition 15%+			-0.105***			-0.044
			(0.026)			(0.035)
Survey: death rate above emergency threshold			0.072*			0.041
			(0.040)			(0.049)
Report: data gap stated			0.008			-0.017
			(0.014)			(0.026)
Area-rounds	8,564	8,564	8,564	8,564	8,564	8,564

Notes: Linear probability models. Sample: FEWS NET areas in Crisis (Phase 3) at map round r, Somalia, Ethiopia and Sudan, 2011–2024, in rounds with at least one extracted report. Outcome: Emergency or worse on the next map (columns 1–3), or by need, the mapped phase plus one if flagged as held back by aid (columns 4–6). "Report" variables are indicators that a FEWS NET report published in the four months up to r said this about the area (extracted with a large language model and matched to mapping units by name). Survey variables refer to survey results quoted in reports, fielded in the 12 months before r. All columns include country-by-round fixed effects. Standard errors clustered by region (first administrative level) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G What a phase means for malnutrition and death

The IPC ties each phase to a band of acute malnutrition among children under five (below 5, 5–10, 10–15, 15–30 and above 30 percent) and of crude death rates (below 0.5 deaths per 10,000 people per day in Phases 1 and 2, 0.5–1 in Crisis, 1–2 in Emergency and above 2 in Famine) (IPC Global Partners, 2021, 2019). The bands were not derived from food insecurity data; they descend from 1980s and 1990s benchmarks for refugee camps (Young and Jaspars, 2009), and the IPC itself notes that food insecurity and acute malnutrition often diverge (IPC Nutrition Working Group, 2016; IPC Global Partnership, 2024). Converting food insecurity into deaths needs a calibration, and a companion paper tests the bands against survey-measured outcomes. We summarize what it finds.

In Somalia, the Food Security and Nutrition Analysis Unit surveys rural livelihood zones twice a year on a fixed calendar, so its surveys are not sent where trouble is suspected. Across 329 zone-seasons, FEWS NET's forecast does not predict measured acute malnutrition (correlation 0.10, against 0.66 for the zone's own survey six months earlier). Within a zone, a forecast one phase worse than usual goes with 0.5–0.8 more points of acute malnutrition and 0.08–0.12 more deaths per 10,000 per day, where the band midpoints imply 5–10 points and 0.45–0.75 deaths. Surveys elsewhere show a steeper gradient for malnutrition, 1.8–3.3 points per phase, and none for crude death rates. The IPC's own malnutrition and food insecurity scales agree for only 35 percent of areas classified on both. And dividing the excess deaths estimated in documented crises by the number of people in Crisis or worse gives about 0.15–0.2 deaths per 10,000 per day in Somalia's 2017–18 and 2022 droughts (Warsame et al., 2023; London School of Hygiene & Tropical Medicine et al., 2023; Ouchtar et al., 2026), about 0.9 in northeast Nigeria in 2016–19 (Checchi et al., 2023), and 1.1–1.9 in the 2010–12 Somalia famine, which met a late response (Checchi and Robinson, 2013).

G.1 A calibration

Table A3 summarizes the evidence and the values we use in Appendix H. For each step from one phase to the next below Famine, we use 0.15 extra deaths per 10,000 per day (range 0.07–0.25) and 1.5 points of acute malnutrition (range 0.6–2.5). For the step from Emergency to Famine, where the evidence comes from famines with a late response, we use 1.5 extra deaths (range 0.9–1.9). The central values are a fifth to a half of what the IPC bands imply for ordinary steps, and the low end of the range comes from Somalia's scheduled surveys, the most selection-free evidence we have.

Table A3: What one phase means: evidence and the values used in the benefit-cost calculation

Evidence	Acute malnutrition (points per phase)	Crude death rate (per 10,000 per day, per phase)
IPC reference bands, midpoints (Stressed to Crisis; Crisis to Emergency)	5; 10	0.45; 0.75
Somalia, scheduled surveys, within zones	0.5–0.8	0.08–0.12
SMART+ surveys, country-by-year effects	1.8	0.00 (FEWS NET); 0.17 (IPC/CH)
Nigeria recurring rounds; Burkina Faso and Mauritania rounds	2.7; 3.3	–
Excess deaths per person in Crisis or worse: Somalia 2017–18, 2022	–	0.15–0.2
Excess deaths per person in Crisis or worse: famines with late response	–	0.9–1.9
Used below Famine (central; range)	1.5; 0.6–2.5	0.15; 0.07–0.25
Used from Emergency to Famine	–	1.5; 0.9–1.9

Notes: Survey estimates are slopes of the measured outcome on the phase; Somalia within-zone estimates with zone effects, with and without season effects. IPC reference bands: differences between band midpoints. Excess deaths per person in Crisis or worse are estimated excess deaths divided by the person-days in IPC Phase 3 or worse; see text for sources.

H Converting food insecurity into deaths

This appendix converts FEWS NET's influence on aid into deaths averted, as a sensitivity analysis. It credits FEWS NET only with the aid that follows its distinctive information, takes FEWS NET's own aid flags at face value as marking where aid holds an area one phase better, and uses the calibration of deaths per phase in Appendix G.

FEWS NET's word on what aid prevents. We take FEWS NET's aid flag at face value: in every area-round it flags, aid keeps the area one phase better than it would otherwise be. Between 2011 and 2024, FEWS NET flagged 11,671 area-rounds. We attach a population to each flagged area using WorldPop's 2020 gridded estimates (WorldPop, 2020), and count the months until the next map. On average, 28 million people lived in a flagged area at any time. Half of the flagged area-rounds (50 percent) were in Stressed held back from Crisis, and 42 percent were in Crisis held back from Emergency.

What a phase means. We use the calibration in Appendix G. Each step from one phase to the next below Famine adds 0.15 deaths per 10,000 people per day (low 0.07, high 0.25) and 1.5 points of acute malnutrition among children under five (low 0.6, high 2.5); the step from Emergency to Famine adds 1.5 deaths (low 0.9, high 1.9). In the official IPC and Cadre Harmonisé data, 10 percent of people in an area classified as Stressed are themselves in Crisis or worse; the figure is 34 percent in Crisis areas and 60 percent in Emergency areas.

On FEWS NET's own account, then, aid in its flagged areas averts 80,000 (low), 168,000 (central) or 274,000 (high) deaths a year, and keeps about 6.5 million people a year out of Crisis or worse. Taking the IPC reference bands literally instead would give 121,000 to 820,000 deaths a year. For scale, the 2011 Somalia famine killed an estimated 258,000 people over two years (Checchi and

Robinson, 2013); the IPC bands imply that aid prevents a disaster of that size every six months, which the evidence in Appendix G does not support.

FEWS NET's share of that aid. Not all of this aid is due to FEWS NET. We credit FEWS NET only with the aid that follows its distinctive information. That is the part of its forecast that the country's usual situation, the season, rainfall, conflict and prices — and, in one version, the latest official analysis — cannot predict (Sections 6 and 7). If funding rises by β log points per unit of this contribution x for a country and round, the share of funding due to FEWS NET is $1-e^{-\beta x}$. Where FEWS NET was less alarmed than other information ($x < 0$), it lowered funding. We then count $-(1-e^{\beta x})$: the funding it subtracted, as a share of what there would have been without it. Both branches measure FEWS NET's effect as a share of the larger of actual and counterfactual funding, so over- and under-alarm count equally. The funding responses β come from Table A1: 10 to 28 percent more funding per 10 percentage points.

This is the right counterfactual for a funding decision. Without FEWS NET, donors would still know that Somalia and South Sudan are chronically in crisis, and when the lean season falls; they would lose what FEWS NET tells them beyond that. Crediting FEWS NET with its whole forecast instead — as if donors would otherwise know nothing — gives $350–1,900 per death averted, but we regard that as an implausible upper bound.

Result. Table A4 and Figure A3 combine the pieces. FEWS NET's distinctive information accounts for 0.3–0.4 percent of the aid in its flagged areas: 0.3 to 0.4 percent of all humanitarian funding, and 0.3 percent of food and nutrition funding. Applied to the aid that FEWS NET says is holding back crises, this gives:

- **Deaths:** 290–970 averted a year, or $46,000–150,000 of FEWS NET's budget per death averted ($36,000–220,000 across the budget range). That is an order of magnitude more than the $3,000–5,500 of the most cost-effective charities (GiveWell, 2024).
- **Hunger:** 41,000–110,000 person-years in Emergency or worse averted a year, at $400–1,100 each, and 17,000–28,000 person-years in Crisis or worse, at $1,600–2,700 each.[1]
- **Child malnutrition:** 50–240 child-years of acute malnutrition averted a year under the central calibration, at $190,000–910,000 each.

The shares are small partly for a mechanical reason. FEWS NET's forecasts assume the aid that is coming, so where aid is flowing — exactly the flagged areas — they tend to sit *below* what other information would predict, and FEWS NET's distinctive information looks like less alarm, not more. This matters most in the half of flagged areas that were Stressed and held back from Crisis. Under the IPC bands' most conservative reading, those areas carry no deaths; under the evidence-based calibration, they carry as many as the areas held back from Emergency, and FEWS

[1]Fewer person-years are kept out of Crisis than out of Emergency because FEWS NET's distinctive information is, on balance, slightly *less* alarmed than other information in Stressed areas held back from Crisis, which pulls down the Crisis total.

NET's less alarmed signal there offsets much of what it adds elsewhere. Applying the one-sided formula $1 - e^{-\beta x}$ to negative signals as well nets out below zero.

The IPC bands as a sensitivity check. With the IPC reference bands instead of the evidence-based calibration, FEWS NET's distinctive information averts 3,100–12,000 deaths a year, at $3,700–15,000 per death. That range is around the cost per death of the most cost-effective charities. But it rests on death rates per phase several times higher than those measured (Appendix G), so we do not treat it as the main estimate.

Table A4: A rough benefit-cost calculation: FEWS NET credited with its distinctive information

FEWS NET credited with its distinctive information on:	Share of aid attributed	Deaths averted per year			Person-years out of Emergency+ per year	Child-years of acute malnutrition per year
		Low	Central	High		
All humanitarian funding; beyond data and official analyses	0.3%	290	460	520	41,000	66–160
All humanitarian funding; beyond data	0.4%	440	750	970	56,000	120–370
Food and nutrition funding; beyond data	0.3%	420	540	380	110,000	16–60
Cost per outcome averted (US$, budget $45 million a year)						
All humanitarian funding; beyond data and official analyses		150,000	99,000	87,000	1,100	270,000–680,000
All humanitarian funding; beyond data		100,000	60,000	46,000	800	120,000–380,000
Food and nutrition funding; beyond data		110,000	84,000	120,000	400	750,000–2,800,000

Notes: Areas FEWS NET flagged as held back by aid, 2011–2024, annual averages; aid assumed to keep each area one phase better. Population: WorldPop 2020. Deaths averted: population × months until the next map × the change in crude death rate from one phase, under the low, central and high calibrations of Table A3: 0.07, 0.15 and 0.25 deaths per 10,000 per day for each step below Famine, and 0.9, 1.5 and 1.9 from Emergency to Famine. Person-years out of Emergency or worse: from the average share of people in Emergency or worse in official IPC and Cadre Harmonisé areas of each phase. Child-years of acute malnutrition: children under five (17 percent of the population) × the change in acute malnutrition prevalence per phase (0.6–2.5 points; range across calibrations). Share of aid attributed: $1 - e^{-\beta x}$ for $x \geq 0$ and $-(1 - e^{\beta x})$ for $x < 0$, where x is FEWS NET's forecast beyond what the country, season, rainfall, conflict and prices (and, in the first row, the latest official analysis) predict, and β the within-country funding response (Table A1); averaged over flagged areas weighted by deaths averted under the central calibration. Costs: $45 million a year divided by each outcome.

The calculation leaves out a good deal in both directions.

- **Ways it may overstate.**
  - It takes FEWS NET's flags at face value, though nearly half of the flags on failed warnings predate the warning, and within Somalia and Sudan the flags track where aid partners work only loosely (Section I).
  - Aid that FEWS NET steers may come from other crises where it would also have saved lives. If the displaced aid would have done half as much good, the costs double.
  - The funding responses are associations, not causal effects, and the within-country evidence for Somalia and Sudan does not show located aid responding to FEWS NET's distinctive information (Section I).
- **Ways it may understate.**
  - FEWS NET's forecasts assume planned aid, which biases its distinctive information downward exactly where aid is flowing.
  - The calibration of deaths per phase leans on Somalia's scheduled surveys, which underrepresent the worst-hit areas, and on phase measures noisy enough to bias the slopes toward zero.

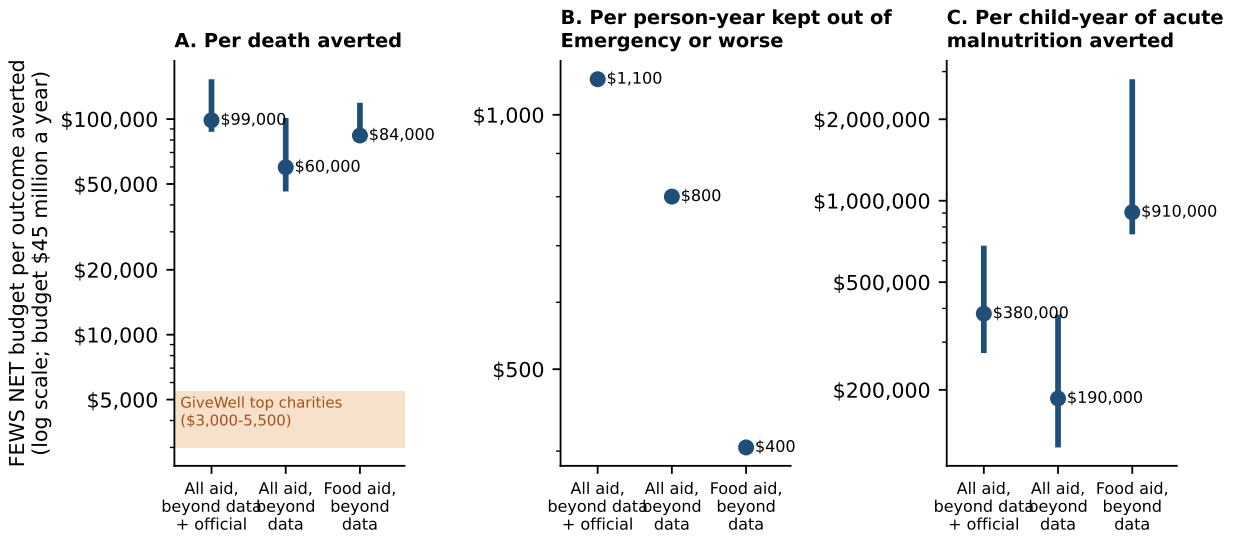


Figure A3: What FEWS NET's budget buys, crediting it only with its distinctive information. Cost per outcome averted, at $45 million a year, for the aid that follows FEWS NET's forecasts beyond what each country's usual situation, the season, rainfall, conflict and prices predict ("beyond data"), or these plus the latest official analysis ("beyond data + official"), using all humanitarian funding or food and nutrition funding. Dots: central calibration of deaths and malnutrition per phase (Appendix G). Bars: range across the low, central and high calibrations. Shaded band: GiveWell's estimate of the cost per death averted by its top charities (GiveWell, 2024).

- It counts only flagged areas. Crises that warnings prevent entirely never show up, nor do responses by governments, markets and households.
- It ignores FEWS NET's value beyond aid allocation, for example to governments, researchers and the official classifications to which its analysts contribute.

With those caveats, the calculation puts FEWS NET's distinctive information well below the most cost-effective global health charities in lives saved: $46,000–150,000 per death averted, against $3,000–5,500. Measured in food insecurity rather than lives, it does better, at about $400–1,100 per person-year kept out of Emergency. The gap between the two is the central lesson of Appendix G: a phase of food insecurity is worth far fewer deaths than the IPC bands suggest.

I Aid and FEWS NET within Somalia and Sudan

Humanitarian funding data are recorded by country, so Section 7 could ask only whether a country's funding moves with FEWS NET's view of the country. The benefit-cost calculation also rests on FEWS NET's district-level judgments of where aid is holding crises back. Neither can be checked without aid data below the country level. Such data exist only from about 2014, and best for Somalia and Sudan. We use three kinds.

- **Money:** allocations from the UN's country-based pooled funds, which record each project's budget by district and sector. They are small, about 3–5 percent of humanitarian funding,

but they are allocated by OCHA with explicit geographic priorities, so they test whether the system's own targeting follows FEWS NET.

- **Presence:** OCHA's records of which organizations work in which district ("Who does What Where"), from which we count food security and nutrition partners.
- **People reached** by food security and nutrition partners, where it is reported in usable form: Somalia in 2022–24 and Sudan in 2019–22.

We match all three to districts (Somalia) and localities (Sudan) by their official codes, and FEWS NET's mapping units to the same districts, giving 74 Somali districts over 33 FEWS NET periods and 192 Sudanese localities over 18 periods. A period runs from one FEWS NET map round to the next.

Framework

Let N_{dt} be the unobserved need in district d in period t. At the map round that opens period t, FEWS NET publishes its current map, its forecast for the period (F_{dt}, the share of the district's population forecast to be in Crisis or worse) and its aid flags. Aid A_{dt} is allocated during the period. The phase observed at the next round is $P_{d,t+1} = N_{d,t+1} - \theta A_{dt}$, and FEWS NET's aid flag is its claim that $N - P$ is at least one phase. We estimate three equations.

Allocation. Does aid follow the forecast?

$$A_{dt} = \beta F_{dt} + \gamma C_{dt} + \rho A_{d,t-1} + \mu_d + \lambda_t + \varepsilon_{dt},$$

where C_{dt} is the current map. District effects μ_d remove persistent differences in need and access, and period effects λ_t remove the size of the country's envelope, so β asks whether a district gets more aid when FEWS NET forecasts it to be worse than usual. With district effects, lagged aid biases the estimates by roughly one over the number of periods (here 18–33); we also estimate the model in first differences, instrumenting the lagged change in aid with aid two periods back. As in Section 7, we split the forecast into the part predictable from the country, the season, rainfall, conflict and prices, and FEWS NET's own contribution. As a placebo, we add the forecast FEWS NET publishes at the *next* round, which allocations made during period t cannot have used.

Flags. Do FEWS NET's aid flags mark where aid went? We regress the share of a district's population under an aid flag on aid, with district and period effects and controls for the current phase. If the flag records aid holding phases down, it should rise when and where aid arrives.

Aid effects. Estimating θ needs aid that varies for reasons unrelated to need. We try a shift-share instrument: a district's share of the pooled fund's food and nutrition allocations in the first two years, times the fund's total allocation in each period, controlling for that share times the national Crisis share. The instrument is weak (first-stage F of 4 in Somalia and 10 in Sudan), and

the estimates are imprecise and, if anything, of the wrong sign, which suggests that larger envelopes coincide with worse conditions in high-share districts. We report them in the replication files but draw no conclusions from them.

Results

Table A5 gives the main estimates. In Somalia, pooled-fund money follows FEWS NET's forecast. A district whose whole population is forecast in Crisis receives 1.10 more in asinh dollars (standard error 0.37) than when none of it is, and is 8 percentage points more likely to receive any (base: 12 percent of district-periods). The estimate barely changes with lagged aid or in first differences. People reached also rise with the forecast (1.69, standard error 0.81). But the money follows the part of the forecast that public information predicts (2.82, standard error 0.68), not FEWS NET's distinctive contribution (−0.01, standard error 0.32). And the placebo fails: the next round's forecast predicts this period's allocations almost as well (0.72, standard error 0.25). Money and forecasts track the same evolving conditions; within Somalia, there is no sign that FEWS NET's own information moves the money. In Sudan, none of the three measures of aid follows the forecast. The number of food security and nutrition partners present does not follow it significantly in either country.

FEWS NET's aid flags partly track where aid is. With district and period effects, each additional food security or nutrition partner in a district goes with a flag share 0.009 higher in Somalia (standard error 0.002) and 0.010 higher in Sudan (0.003). In Somalia, a one-standard-deviation rise in people reached per resident goes with a flag share 17 percentage points higher, against a mean of 20 percent. In Sudan, people reached does not predict the flag within localities. Pooled-fund money barely predicts it in either country. The flags are also sticky: in Sudan, 0.72 of a locality's flag share carries over to the next round. The flag is consistent with a standing judgment that aid is present, more than with a response to new aid.

These results qualify two parts of the paper. First, the country-level finding that money follows FEWS NET's distinctive information (Section 7) does not carry down to districts in the two countries we can examine: within Somalia the money that is located follows conditions everyone can see. Pooled funds are a small and atypical slice of aid, so this does not rule out a district-level response by the large agencies whose spending is not located. Second, the aid flags on which the benefit-cost calculation rests mark aid presence only loosely. Both point in the same direction as Appendix G: the case for FEWS NET in lives saved is weaker than its maps alone suggest.

Table A5: Aid and FEWS NET within Somalia and Sudan

Outcome	Regressor	Somalia	Sudan
Pooled-fund aid (asinh US$)	Forecast share in Crisis+	1.099***	-0.325
		(0.370)	(0.376)
	with lagged aid	1.104***	-0.501
		(0.372)	(0.379)
	first differences, lag instrumented	1.424***	-0.779*
		(0.469)	(0.446)
	Predictable part of forecast	2.816***	1.056
		(0.683)	(0.653)
	FEWS NET's distinctive information	-0.006	0.017
		(0.324)	(0.287)
	Placebo: next round's forecast	0.723***	0.199
		(0.251)	(0.279)
Any pooled-fund aid	Forecast share in Crisis+	0.080***	-0.028
		(0.028)	(0.030)
Food and nutrition partners present	Forecast share in Crisis+	0.204	0.386
		(0.327)	(0.396)
People reached (asinh, annual)	Forecast share in Crisis+	1.688**	-0.205
		(0.805)	(0.383)
Aid flag share	Pooled-fund aid (asinh)	0.004**	-0.002*
		(0.002)	(0.001)
	Partners present	0.009***	0.010***
		(0.002)	(0.003)
	People reached (asinh, annual)	0.012	-0.000
		(0.021)	(0.002)
	Placebo: next period's aid	-0.002	-0.002*
		(0.002)	(0.001)
Districts		74	192
FEWS NET periods		33	18

Notes: District (Somalia) or locality (Sudan) by FEWS NET period, 2014–2024 (Sudan from December 2018). All regressions include district and period effects (year effects in the annual regressions for people reached) and control for the share of the district mapped in Crisis or worse; flag regressions also control for the mean phase. Pooled-fund aid: food security, nutrition and multipurpose cash budgets of projects starting in the period (OCHA country-based pooled funds). Partners present: food security and nutrition organizations in OCHA's Who does What Where records. People reached: Somalia 2022–24 (Cash Working Group and OCHA records), Sudan 2019–22 (response plan monitoring). Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Replication

All data are public. The analysis is in Python, in the project's `code` directory.

- FEWS NET data: `01_fetch_fewsnet.py`, `07_build_panel.py`
- Official classifications: `04_fetch_ipc_ch.py`, `23_outcome_panel.py`
- Funding: `03_fetch_fts.py`
- Rainfall, conflict and prices: `22_fetch_chirps.py`, `24_area_rainfall.py`, `26_fetch_acled_wfp.py`
- Type I and II errors: `12_errors.py`; need-defined emergencies: `30_need_onsets.py`
- The 2025 shutdown: `28_flag_release.py`; coverage changes: `27_shock_index.py`
- Funding responses: `31_aid_response.py`
- FEWS NET's distinctive information: `32_forecast_value.py`, `33_fewsnet_contribution.py`
- Official projections: `34_official_projections.py`
- FEWS NET's reports: `50_fetch_reports.py`, `51_extraction_instructions.md`, `52_text_features.py`, `53_text_value.py`; prices: `54_fetch_fdw_prices.py`, `55_price_value.py`; Somalia's scheduled surveys: `56_fsnau_surveys.py`, `57_fsnau_value.py`
- Numbers, figures and tables in this paper: `40_paper.py`, `41_paper_tables.py`, `42_paper_bootstrap.py`